

Lab 4

ELECENG 3EY4

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Objective 1: Review of Lab Manual

Contribution

Objective 2: Alaa

Objective 3: Tina

Objective 4: Marisa

Objective 5: Alaa, Tina, Marisa

Objective 6: Alaa, Tina, Marisa

Objective 3: Test and Report the Motor Parameters

Question 1

There were several motor parameters that were calculated by VESC that represent the key characteristics of the motor and are essential to control the motor. These parameters are displayed in Figure 1 below.

R: “R” represents the motor resistance. The motor resistance parameter shows a value of $4.64\text{m}\Omega$, which indicates the opposition to the flow of current in the motor windings. This changes how much current will flow through the motor windings for a specific voltage. This is essential to control the motor as this affects the current and efficiency of the motor.

L: “L” represents the motor inductance. The motor inductance parameter shows a value of 2.22uH , which reflects the motor winding’s ability to store energy in a magnetic field while current is flowing through it. When there are changes in current, the inductance influences the motor’s response. This is essential to control the motor as this affects the motor’s response to alterations in the current and voltage as well as its efficient operation.

Lq-Ld: “Lq-Ld” represents the motor inductance difference between the q- and d-axis. The motor inductance difference parameter shows a value of 0.33uH , which shows a variation in the motor's inductance that occurs because of winding configuration changes. This is essential to accurately control the motor’s operation.

λ : “ λ ” is lambda, which represents the motor flux linkage. The motor flux linkage (lambda) parameter shows a value of 0.818mWb , which is the total amount of magnetic flux produced by the motor’s magnetic field. This is linked with the motor windings. This is essential to control the motor as this affects the motor's torque production and speed.

KP: “KP” represents the current proportional gain in the current control loop of the motor. The current KP parameter shows a value of 0.0022 . This is essential to control the motor as this controls the amount of control effort required depending on the current error. This is essential in

order to achieve the desired performance and stability of the motor. KP is also an important PID speed controller parameter.

KI: “KI” represents the current integral gain in the current control loop of the motor. The current KI parameter shows a value of 4.64. This is essential to control the motor as this integrates past errors in order to manipulate the control effort over a period of time. This is essential in order to achieve the desired performance and stability of the motor. KI is also an important PID speed controller parameter.

Observer Gain (x1m): “x1m: represents the observer gain. The observer gain parameter shows a value of 1494.49, which is the gain of the observer used to estimate unmeasurable states of the motor. This may include the speed or position of the rotor. Ensuring precise control of the motor is crucial as it influences the accuracy of state estimation.

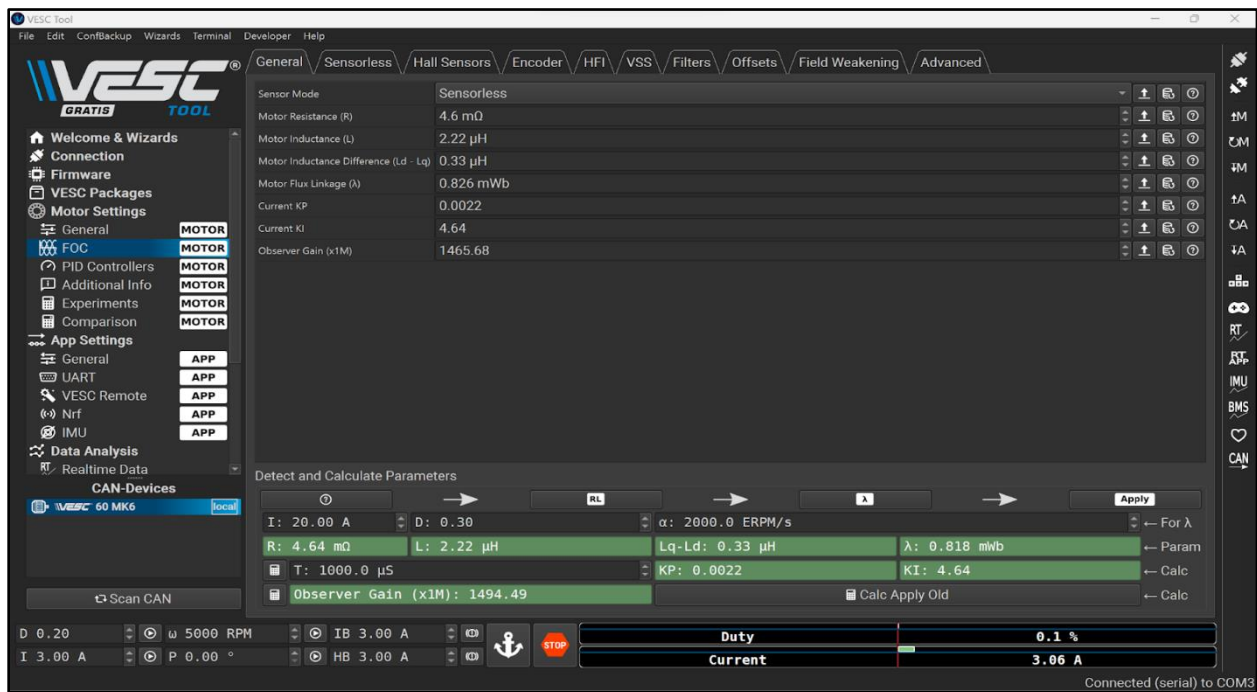


Figure 1. VESC Tool: FOC General Tab Displaying Identification Process Completed and Motor Parameters

Objective 4: Record Real-Time Data and Set PID Parameters

Question 2

The PID speed control parameters are significant in relation to the operation of the McMaster AEV. Changing the PID speed controller parameters as $K_p = 0.008$, $K_i = 0.00225$, and $K_d = 0.00035$ is important because these parameters play a vital role in the operation of the McMaster AEV by regulating its speed, stability, and optimal performance of the vehicle by ensuring smooth and accurate motion.

The current proportional gain, $K_p = 0.008$, determines the proportional response to the error signal between the desired and actual speed. This means that as we increase the K_p value, it amplifies the correction to the error, which thus increases the system's responsiveness. If the K_p value is set too high, it may result in instability.

The current integral gain, $K_i = 0.00225$, controls the integral response to the accumulated error over time. As the K_i value increases, it results in faster correction of steady-state errors, but may lead to overshoot or oscillations if the value is excessively high.

The derivative gain, $K_d = 0.00035$, takes care of the rate of change of the error signal. Increasing the K_d value helps dampen oscillations and improve stability by anticipating future errors based on the current rate of change.

Overall, these PID speed control parameters are significant for the McMaster AEV's speed, correction to disturbances and errors, and stability.

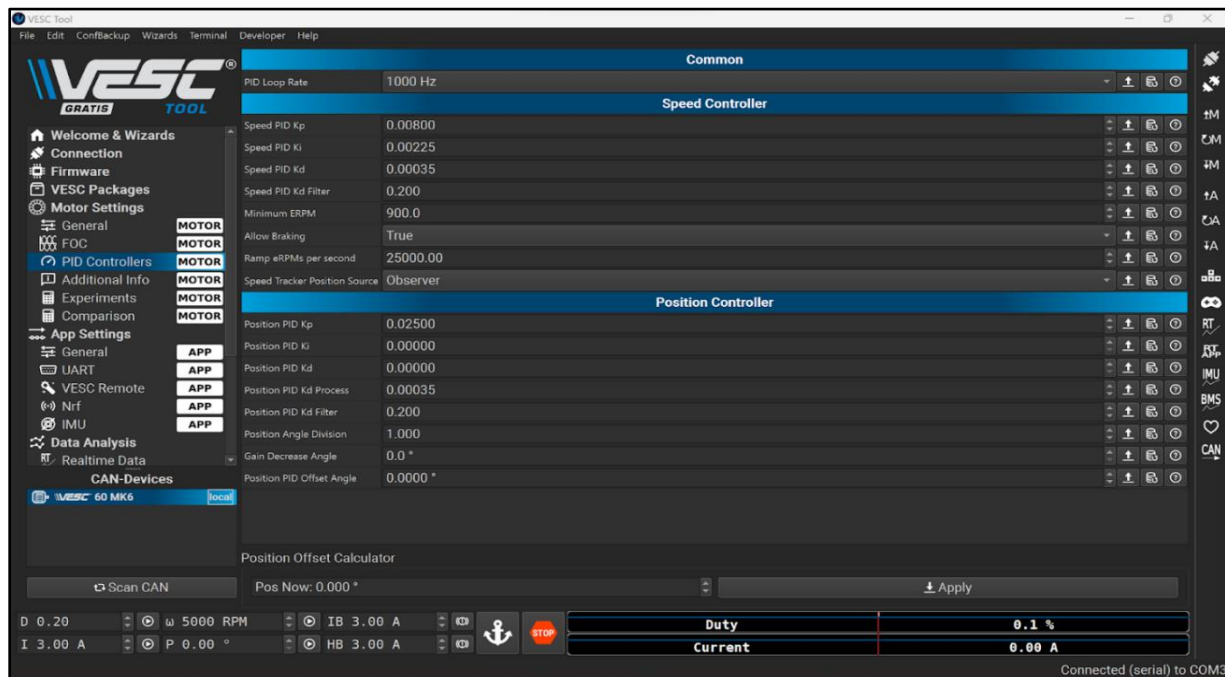


Figure 2. PID Controllers: Changing the Speed Gain Parameters

Question 3



Figure 3. 2500rpm: Current Graph

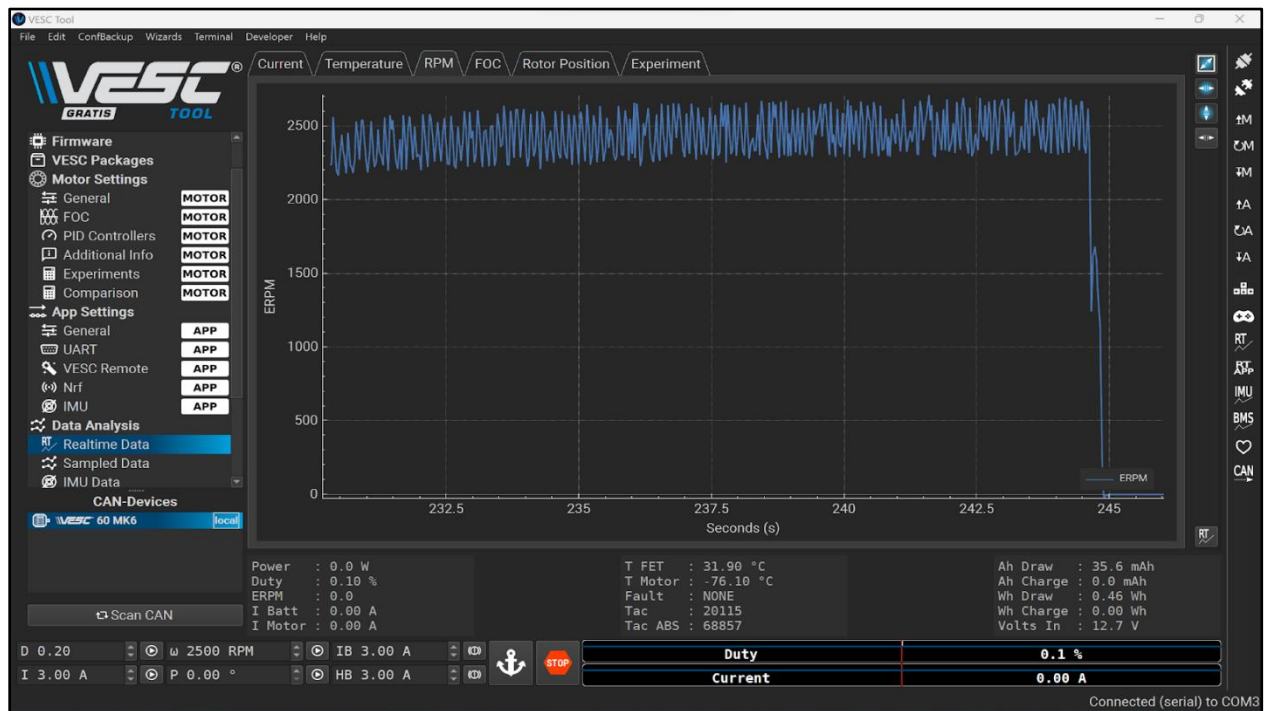


Figure 4. 2500rpm: RPM Graph

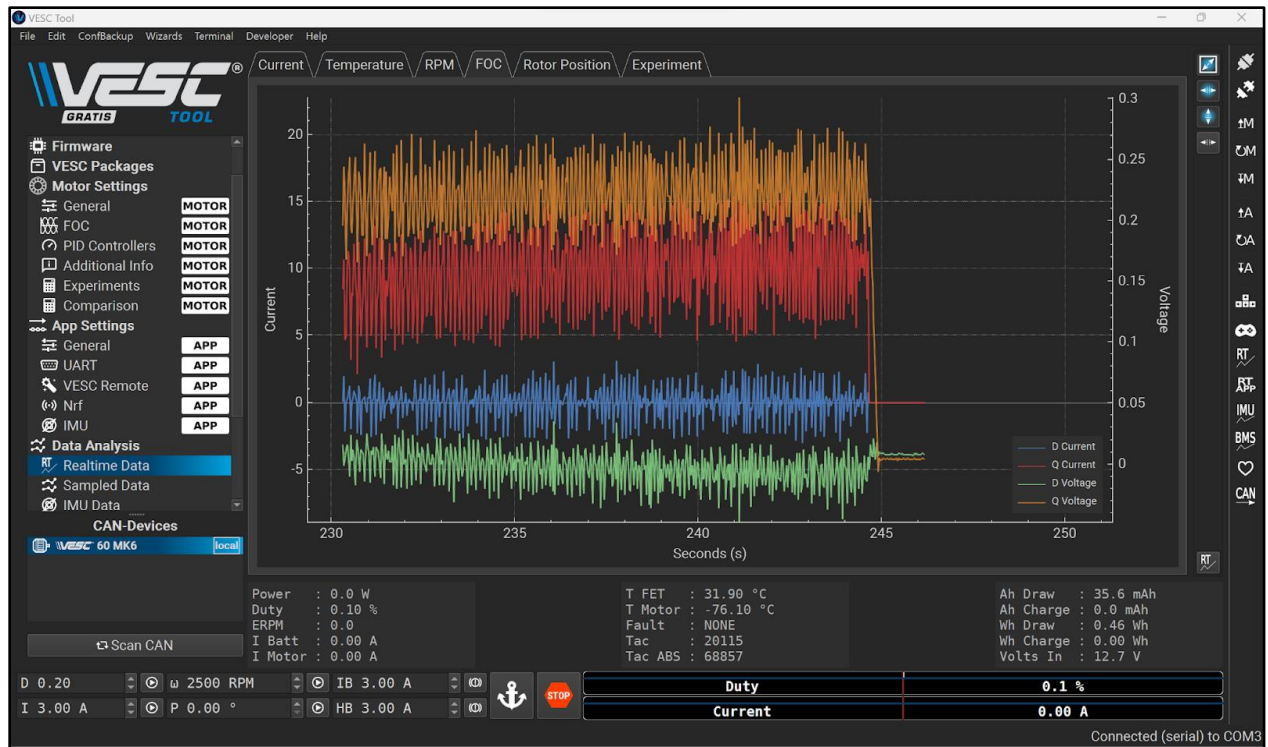


Figure 5. 2500rpm: FOC Graph

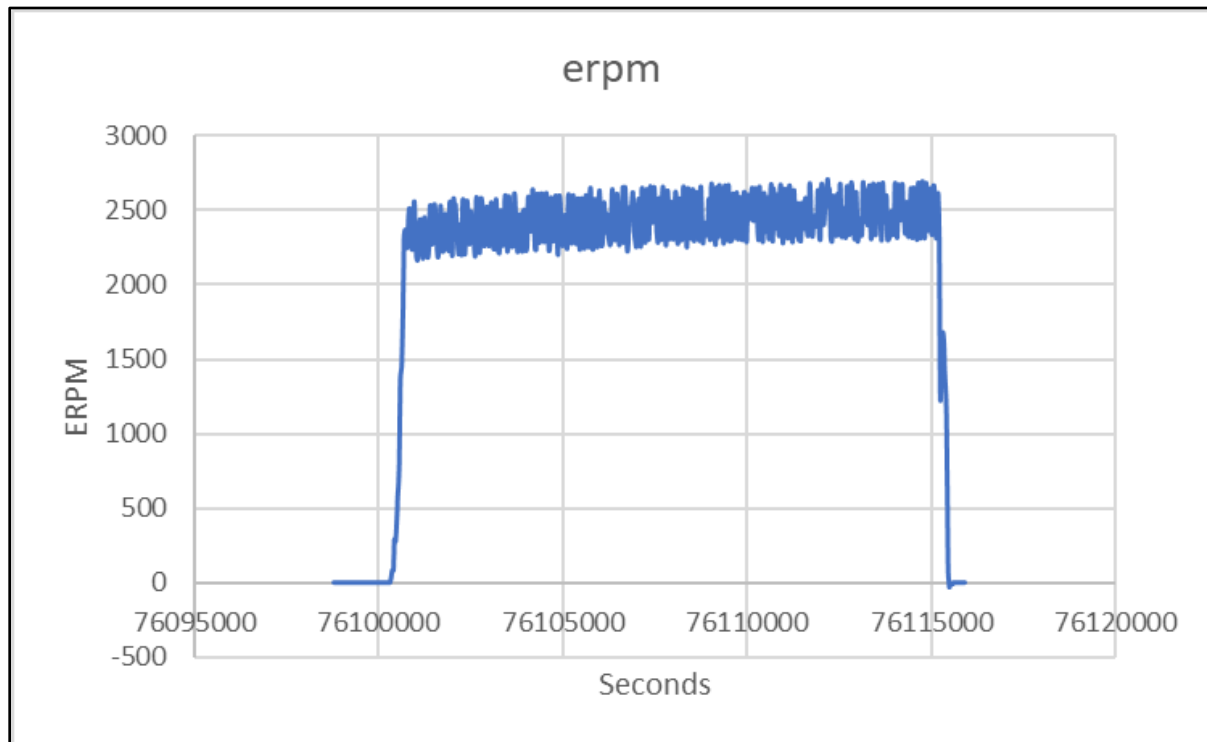


Figure 6. 2500rpm: Excel RPM Graph

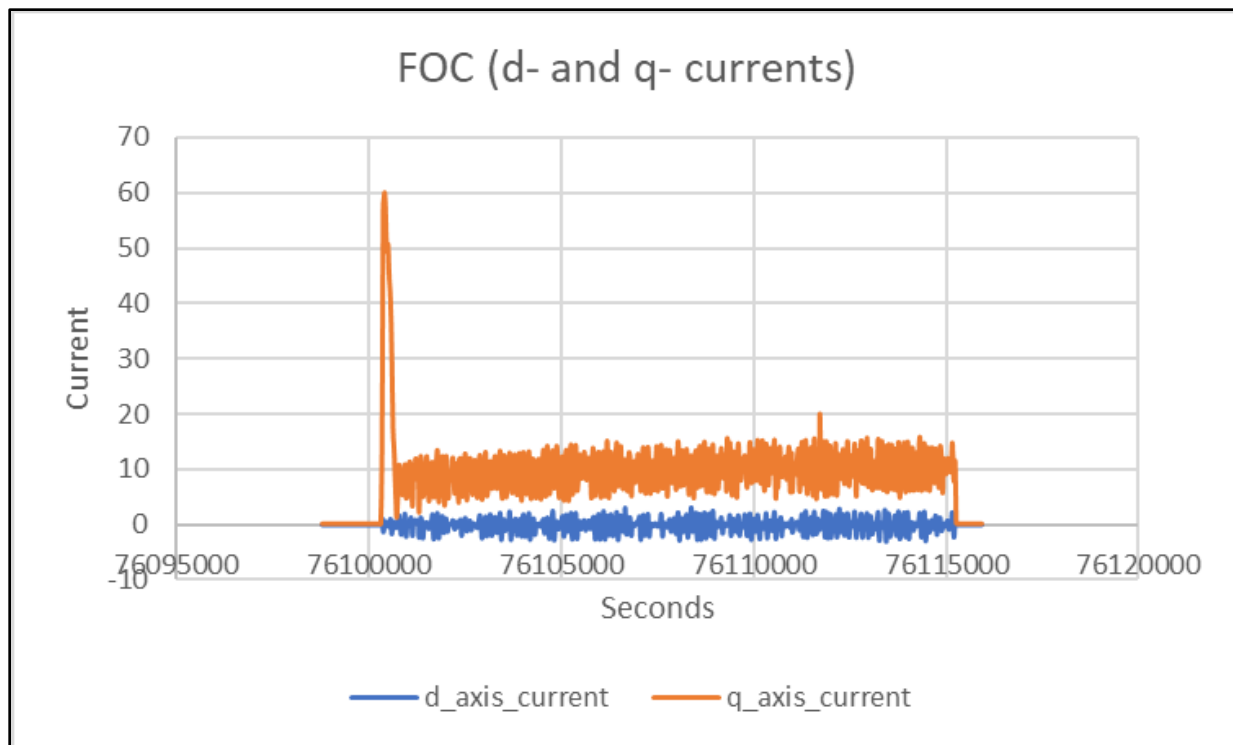


Figure 7. 2500rpm: Excel FOC Current (d- and q-axis Currents) Graph

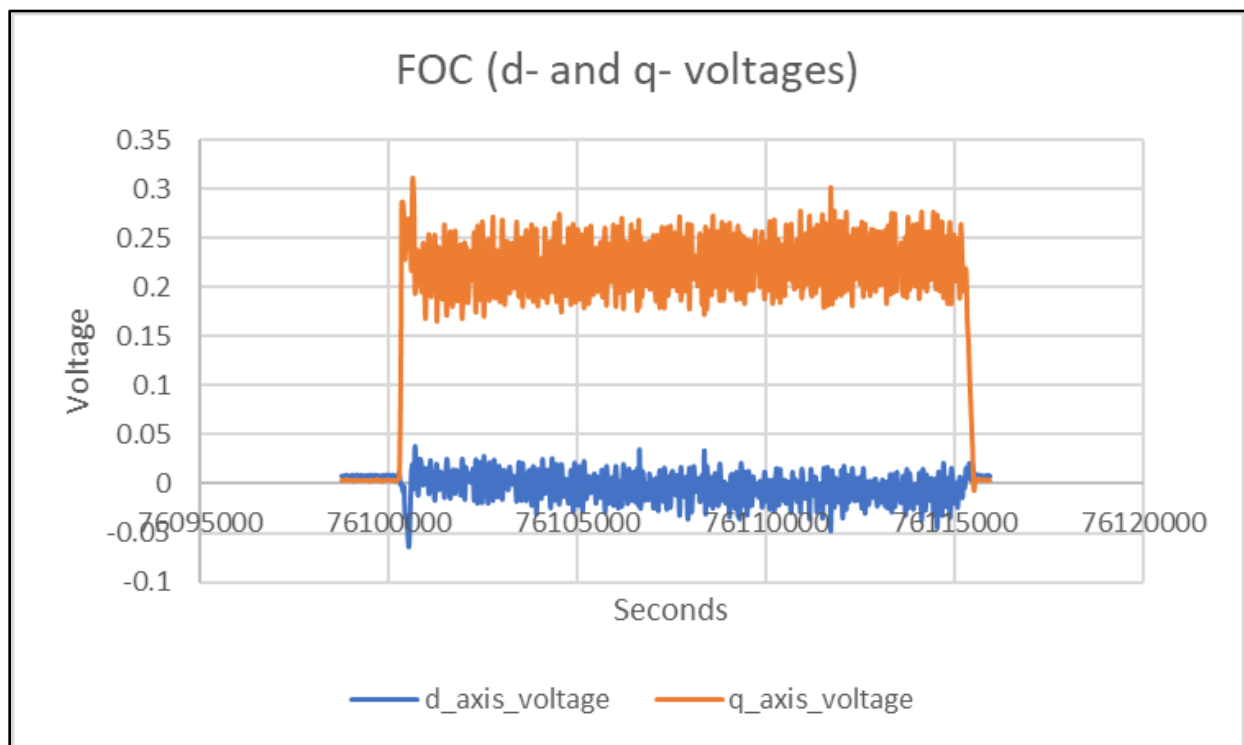


Figure 8. 2500rpm: Excel FOC Voltage (d- and q-axis Voltages) Graph

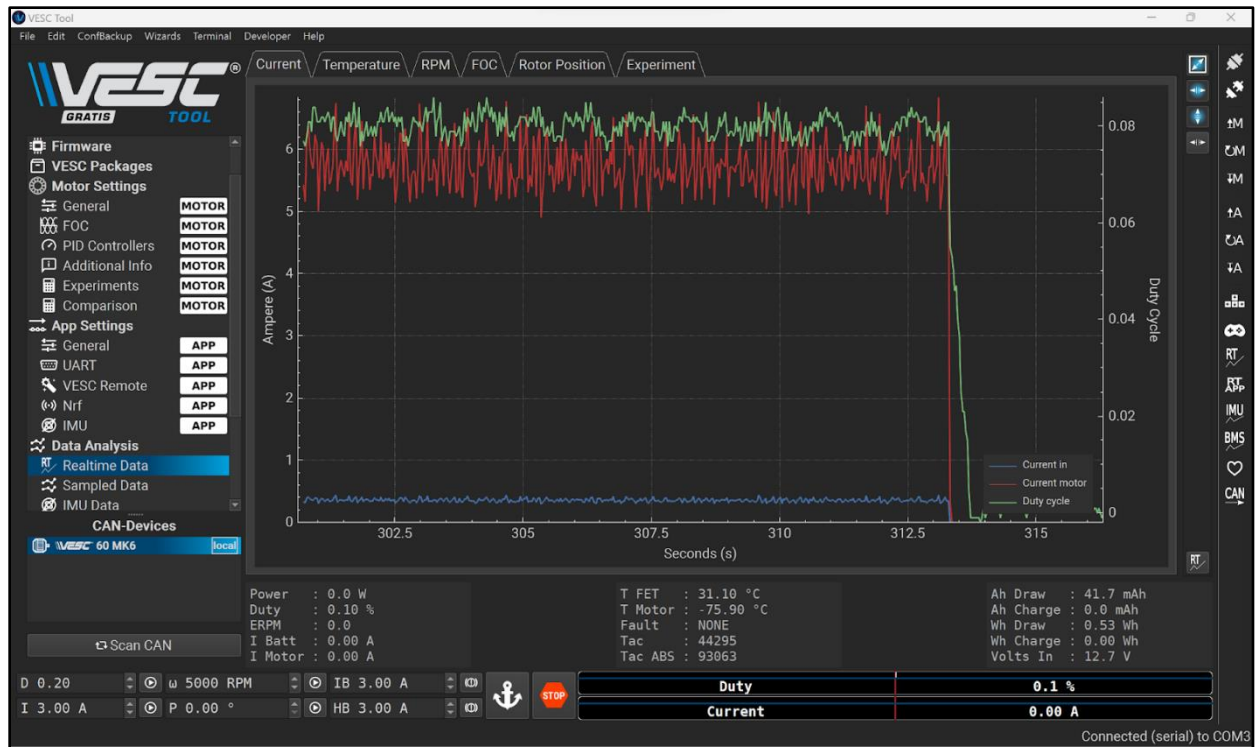


Figure 9. 5000rpm: Current Graph

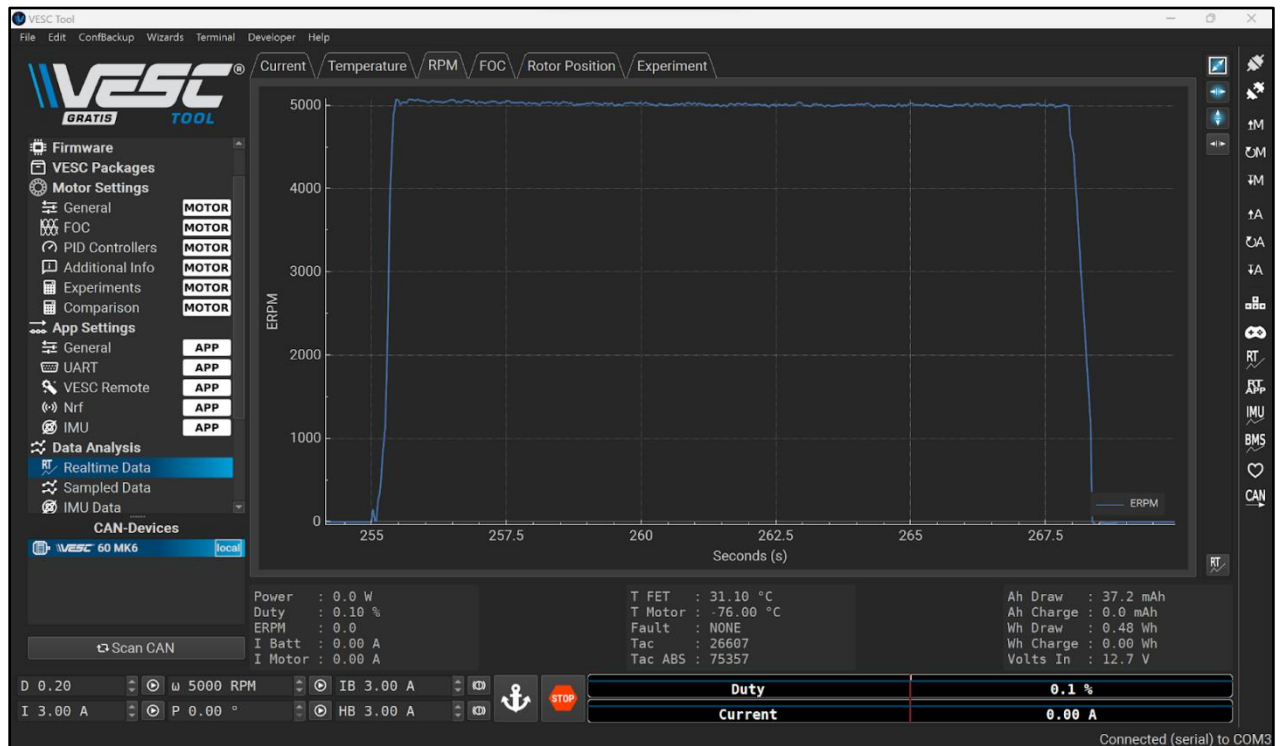


Figure 10. 5000rpm: RPM Graph



Figure 11. 5000rpm: FOC Graph

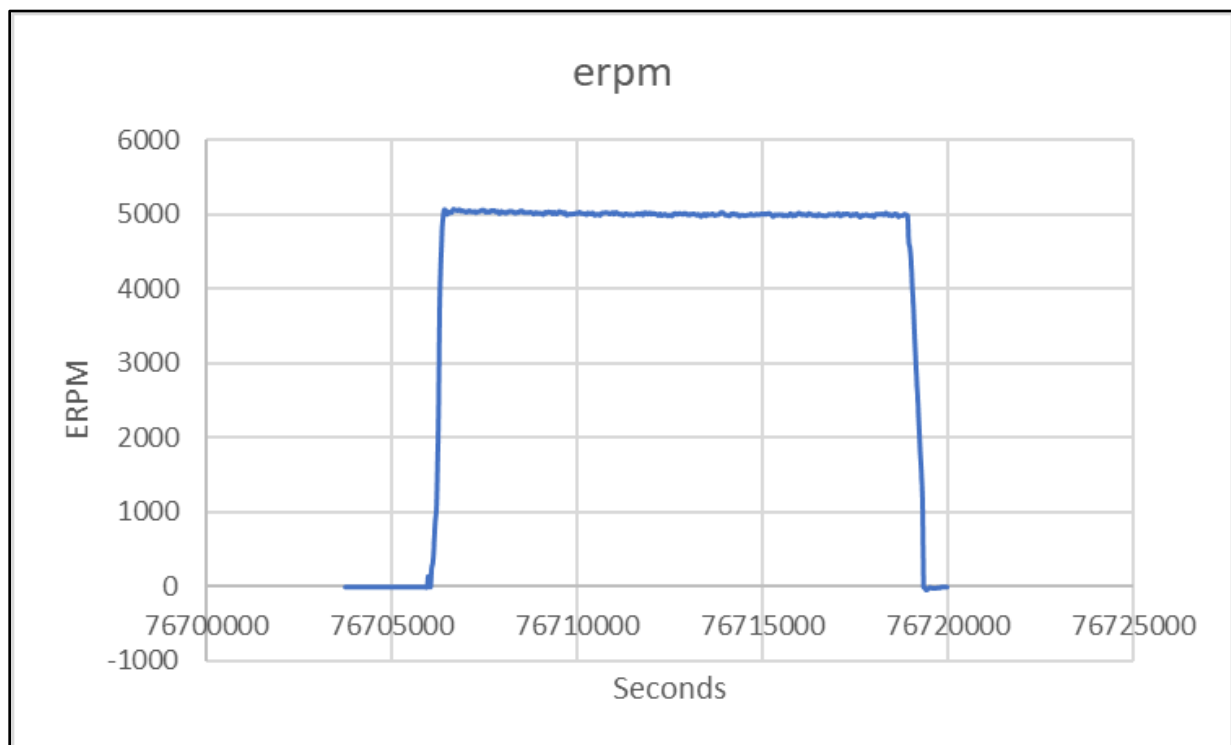


Figure 12. 5000rpm: Excel RPM Graph

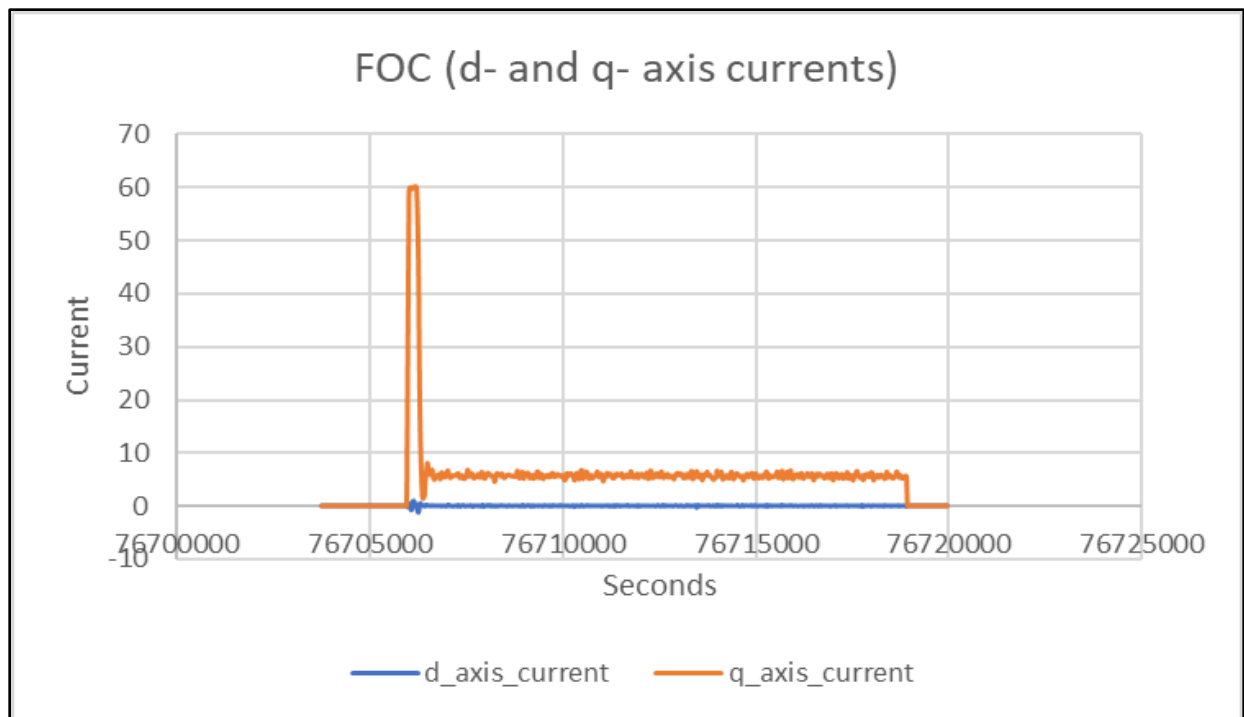


Figure 13. 5000rpm: Excel FOC Current (d- and q-axis Currents) Graph

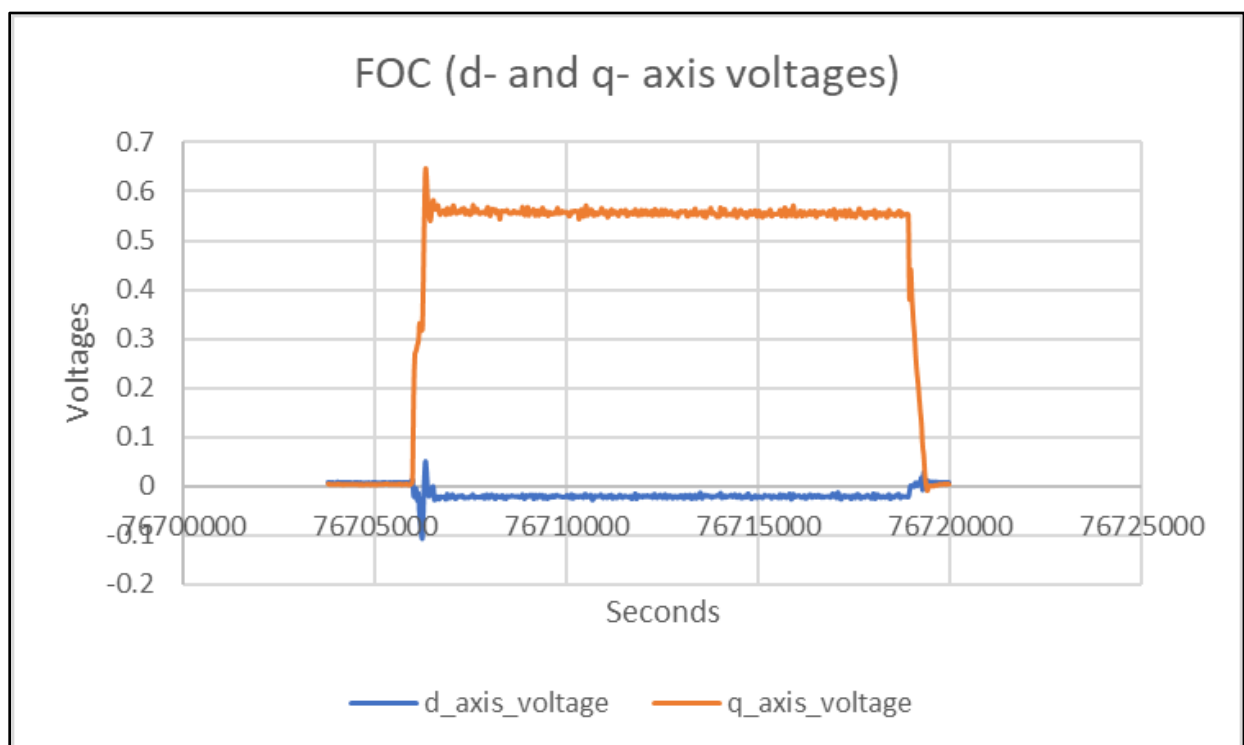


Figure 14. 5000rpm: Excel FOC Voltage (d- and q-axis Voltages) Graph



Figure 15. 7500rpm: Current Graph

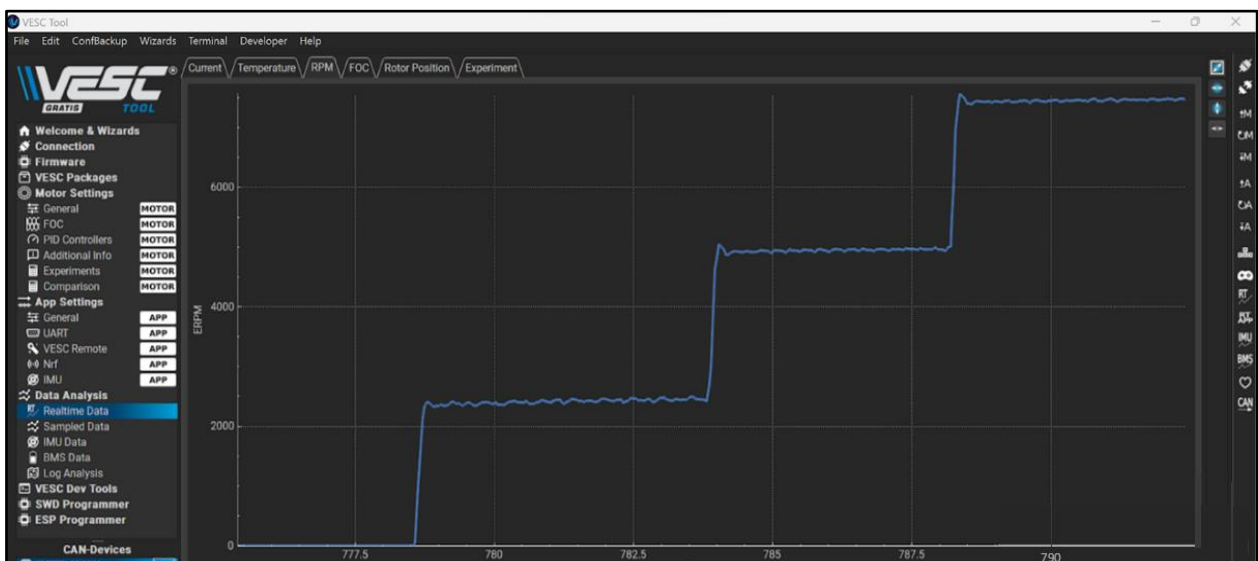


Figure 16. 7500rpm: RPM Graph



Figure 17. 7500rpm: FOC Graph

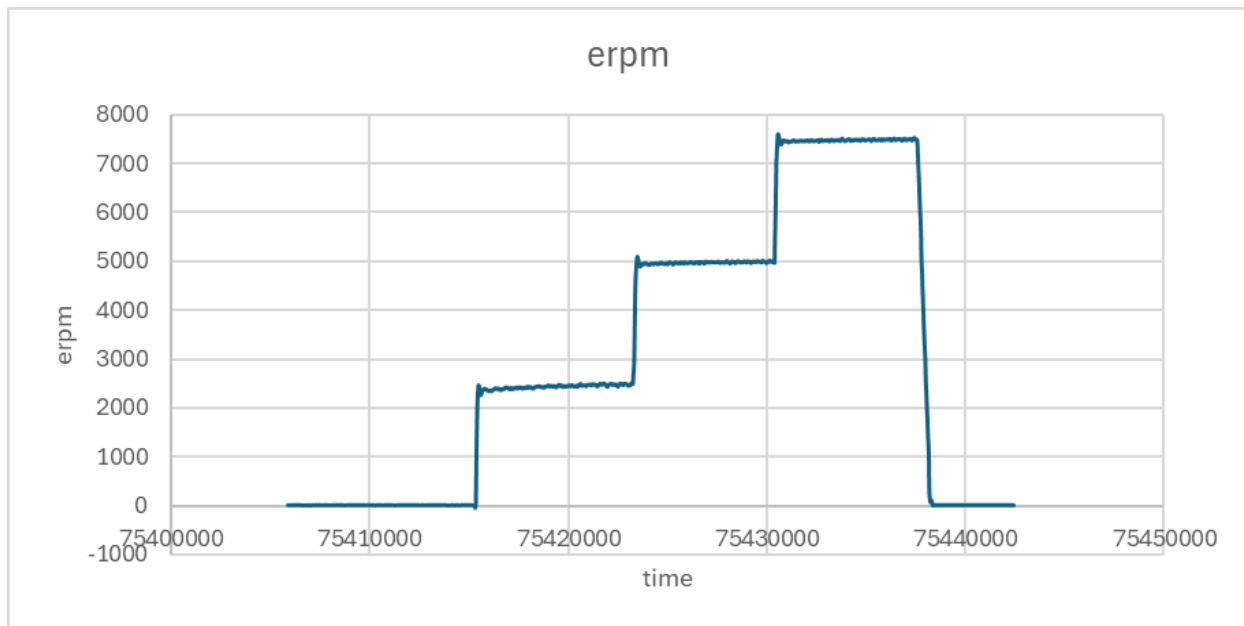


Figure 18. 7500rpm: Excel RPM Graph

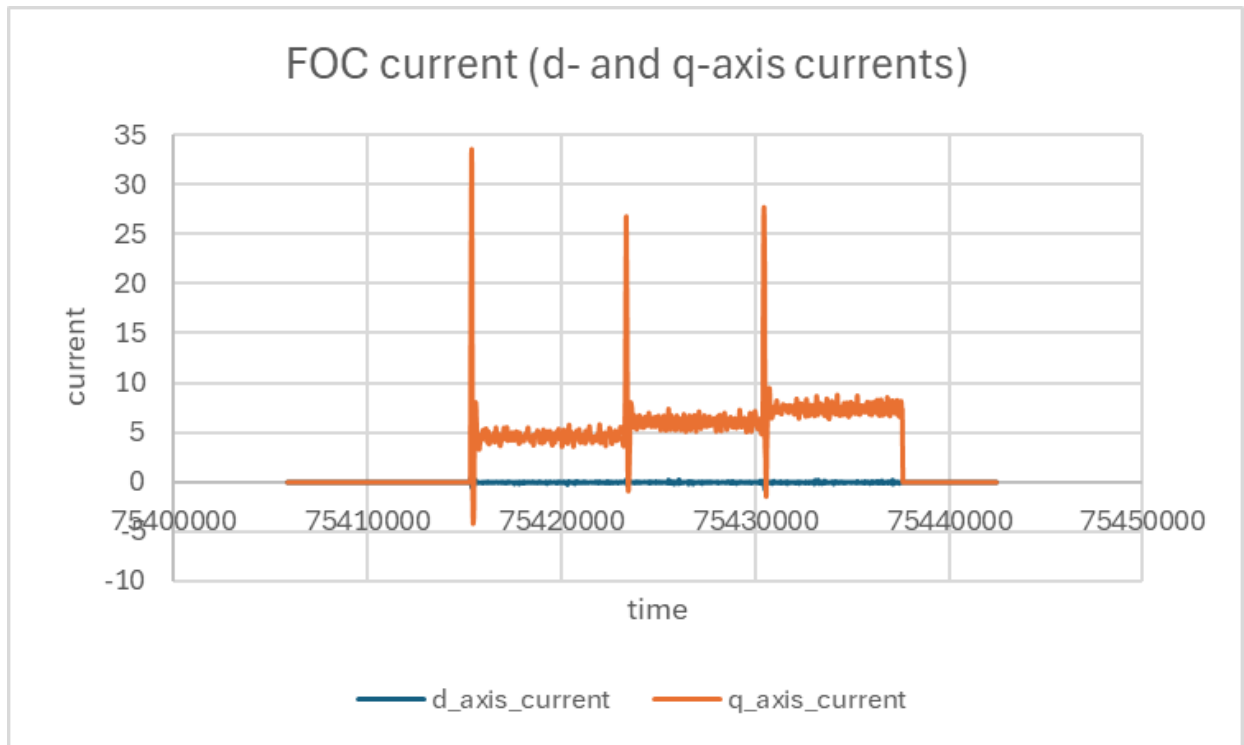


Figure 19. 7500rpm: Excel FOC Current (d- and q-axis Currents) Graph

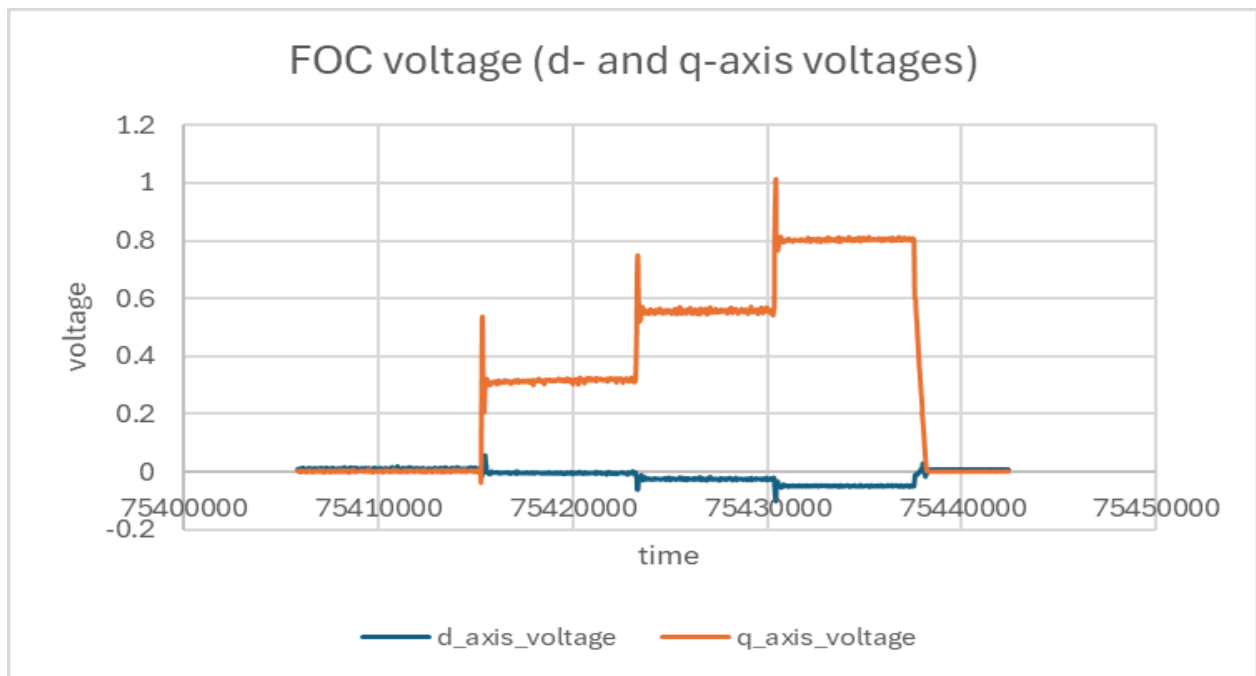


Figure 20. 7500rpm: Excel FOC Voltage (d- and q-axis Voltages) Graph

Question 4

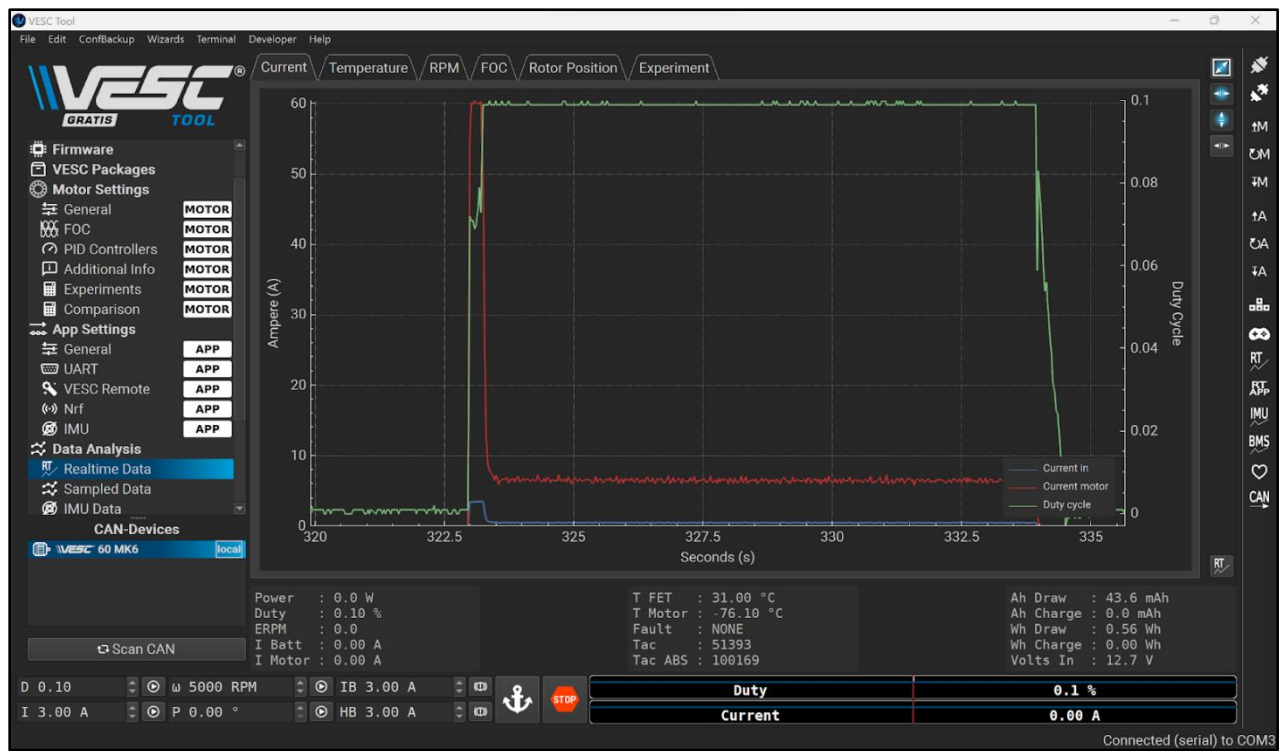


Figure 21. 0.10 Duty Cycle: Current Graph

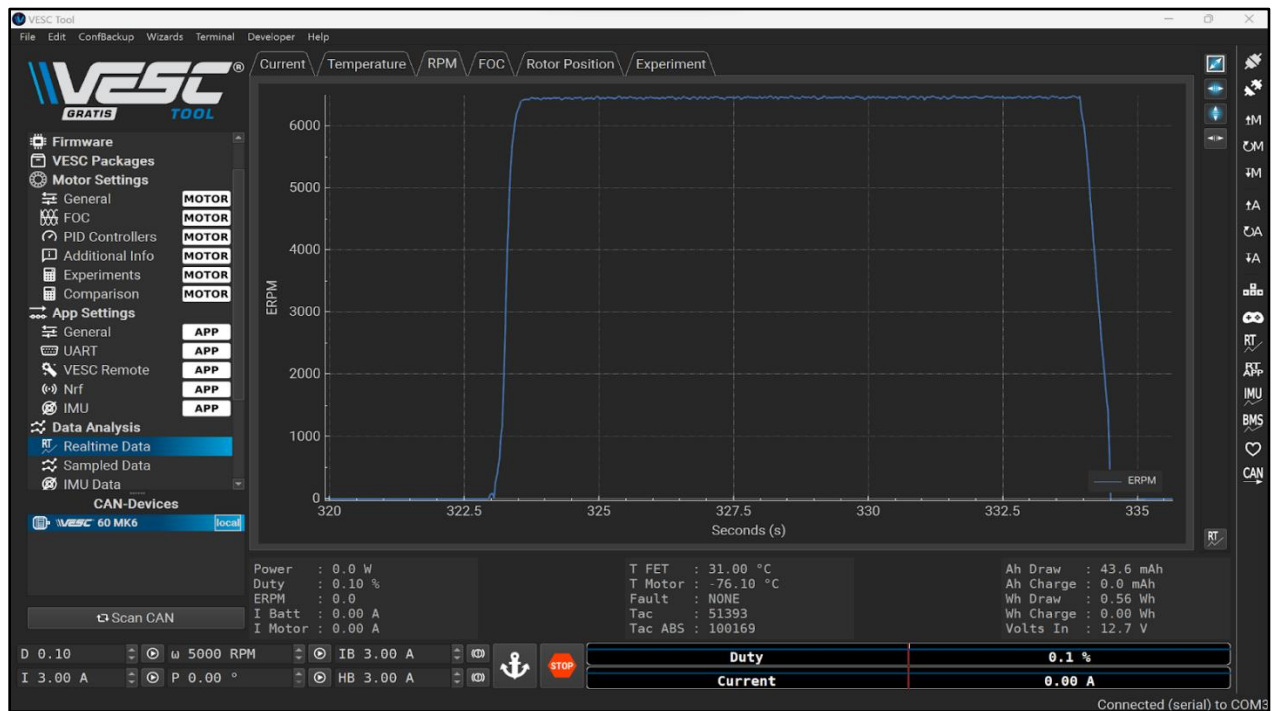


Figure 22. 0.10 Duty Cycle: RPM Graph

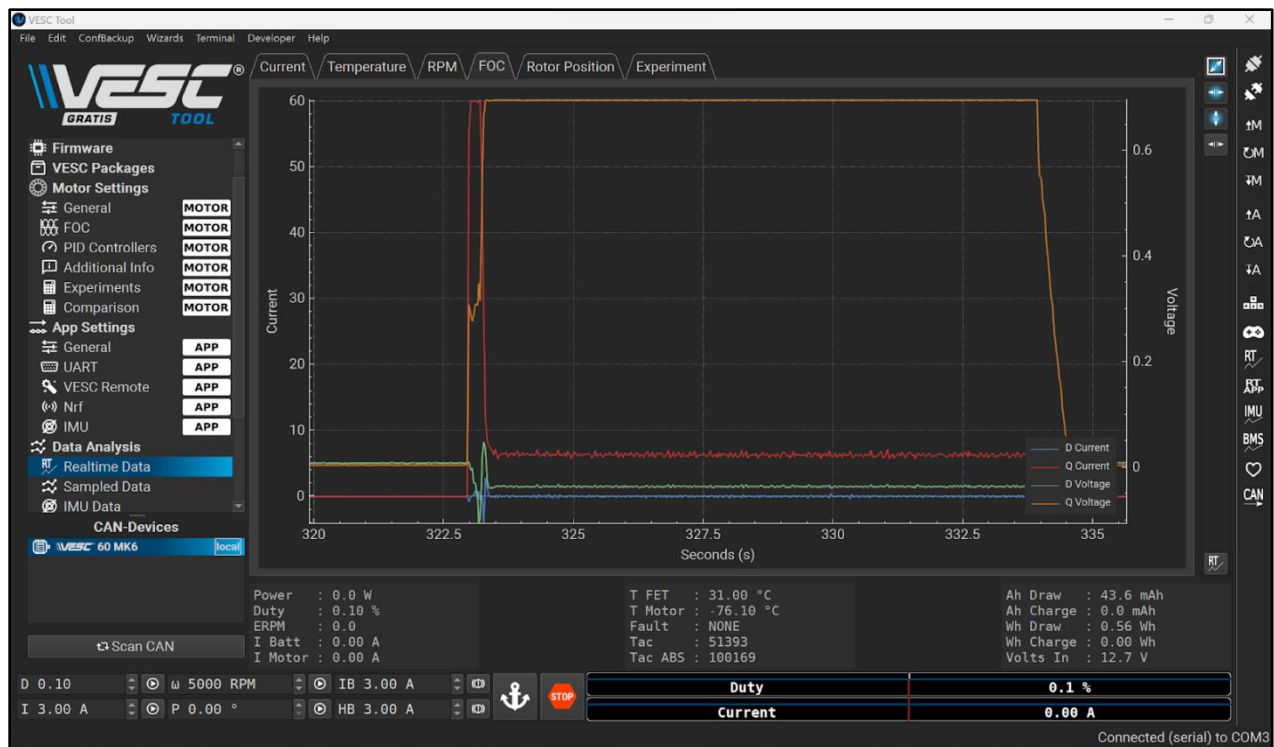


Figure 23. 0.10 Duty Cycle: FOC Graph

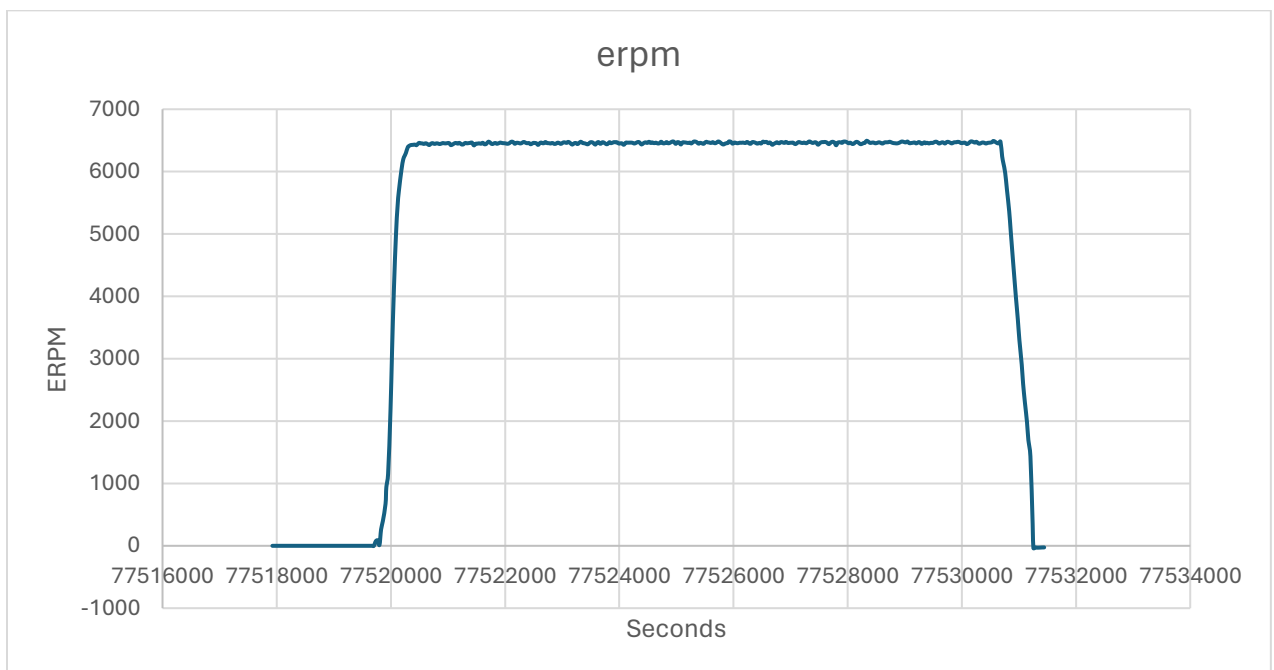


Figure 24. 0.10 Duty Cycle: Excel RPM Graph

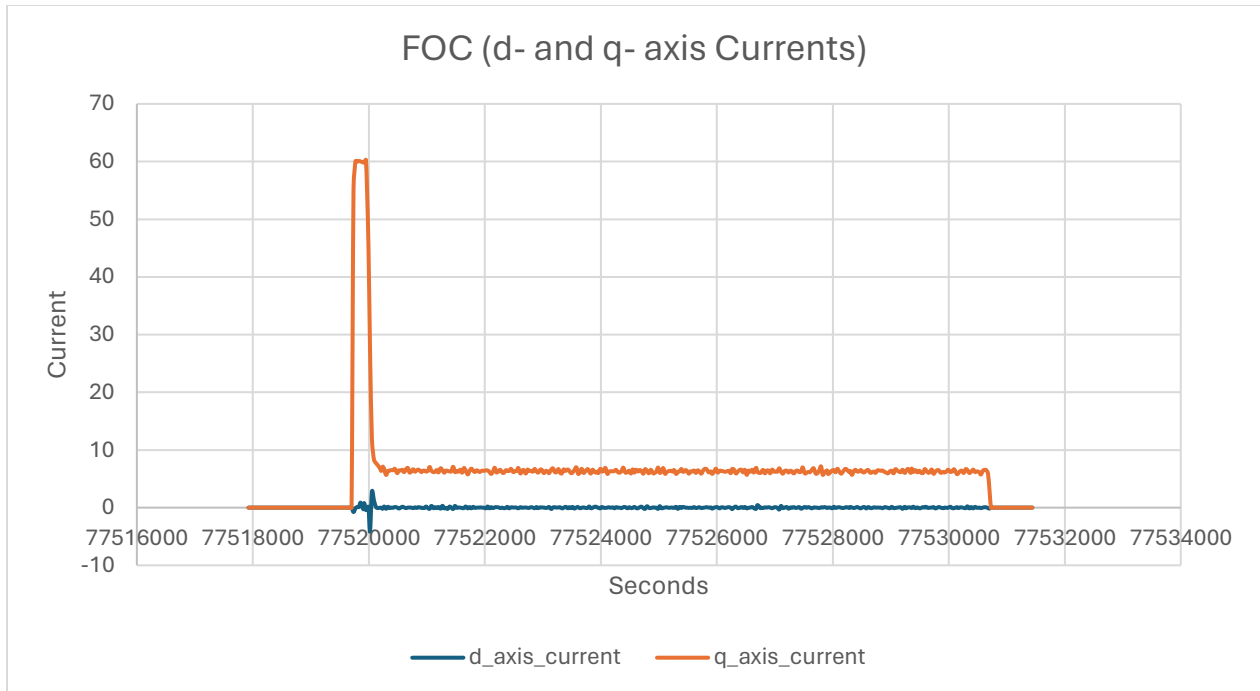


Figure 25. 0.10 Duty Cycle: Excel FOC Current (d- and q-axis Currents) Graph

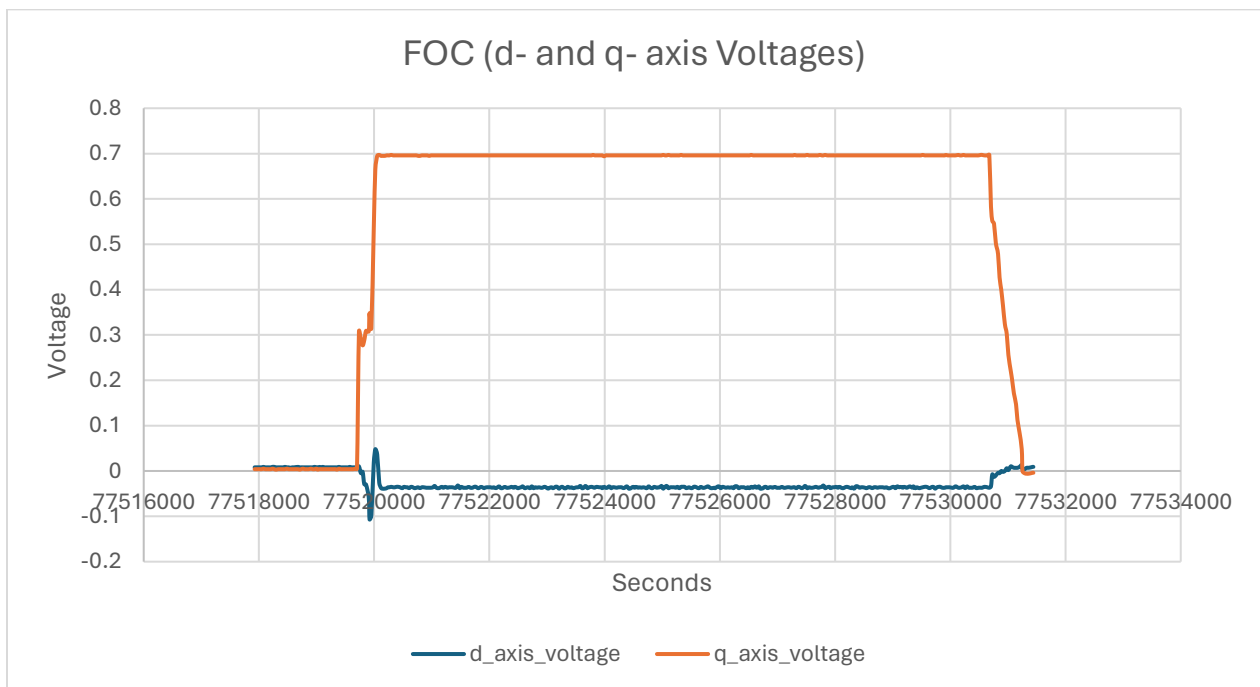


Figure 26. 0.10 Duty Cycle: Excel FOC Voltage (d- and q-axis Voltages) Graph

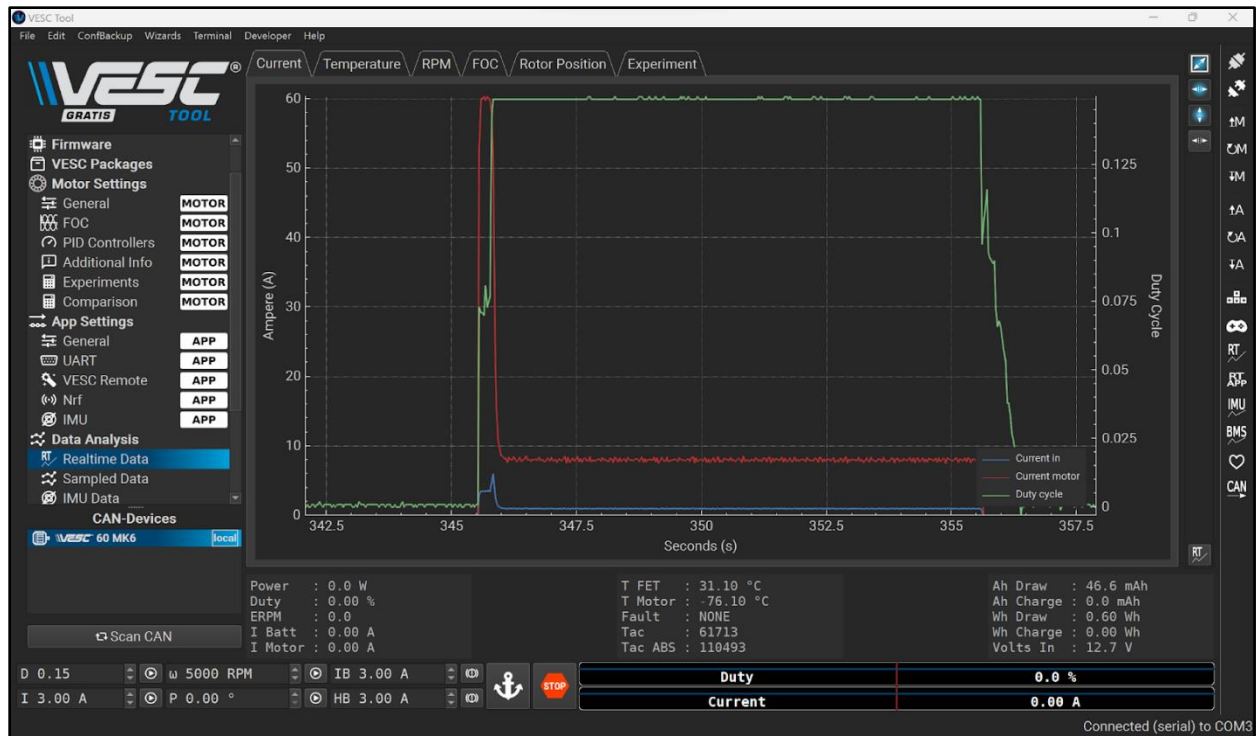


Figure 27. 0.15 Duty Cycle: Current Graph

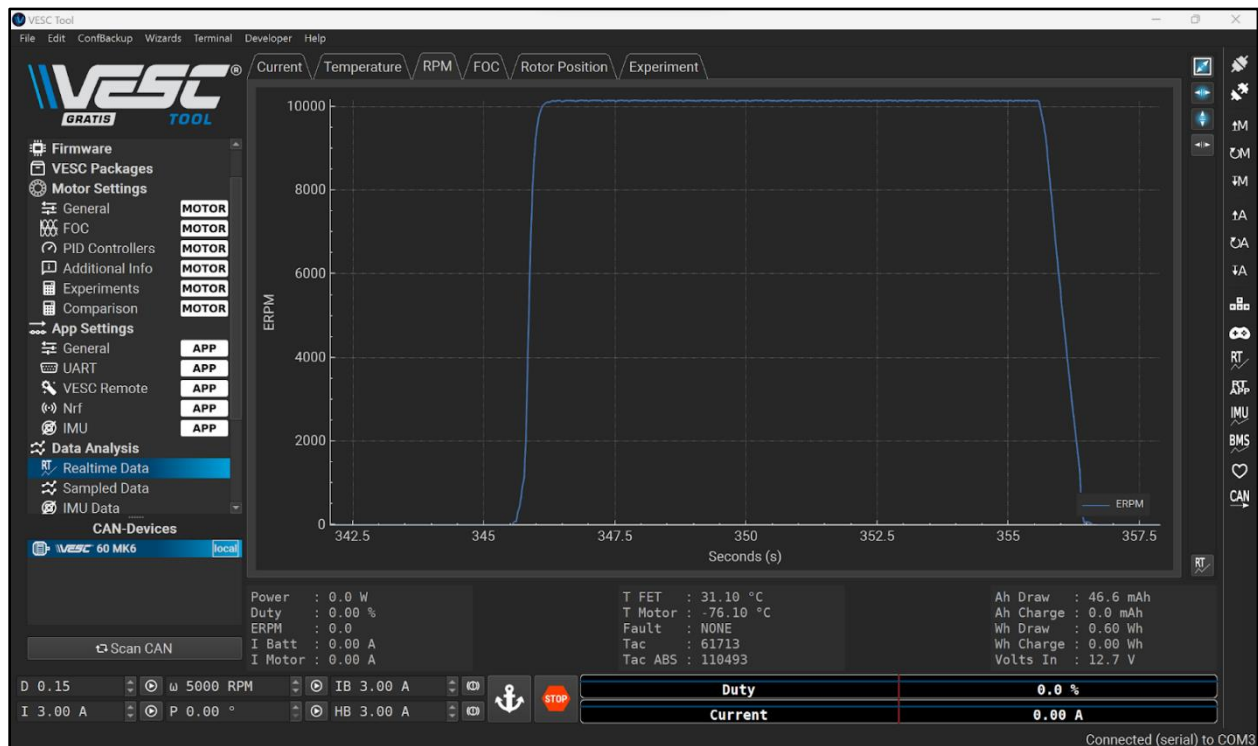


Figure 28. 0.15 Duty Cycle: RPM Graph

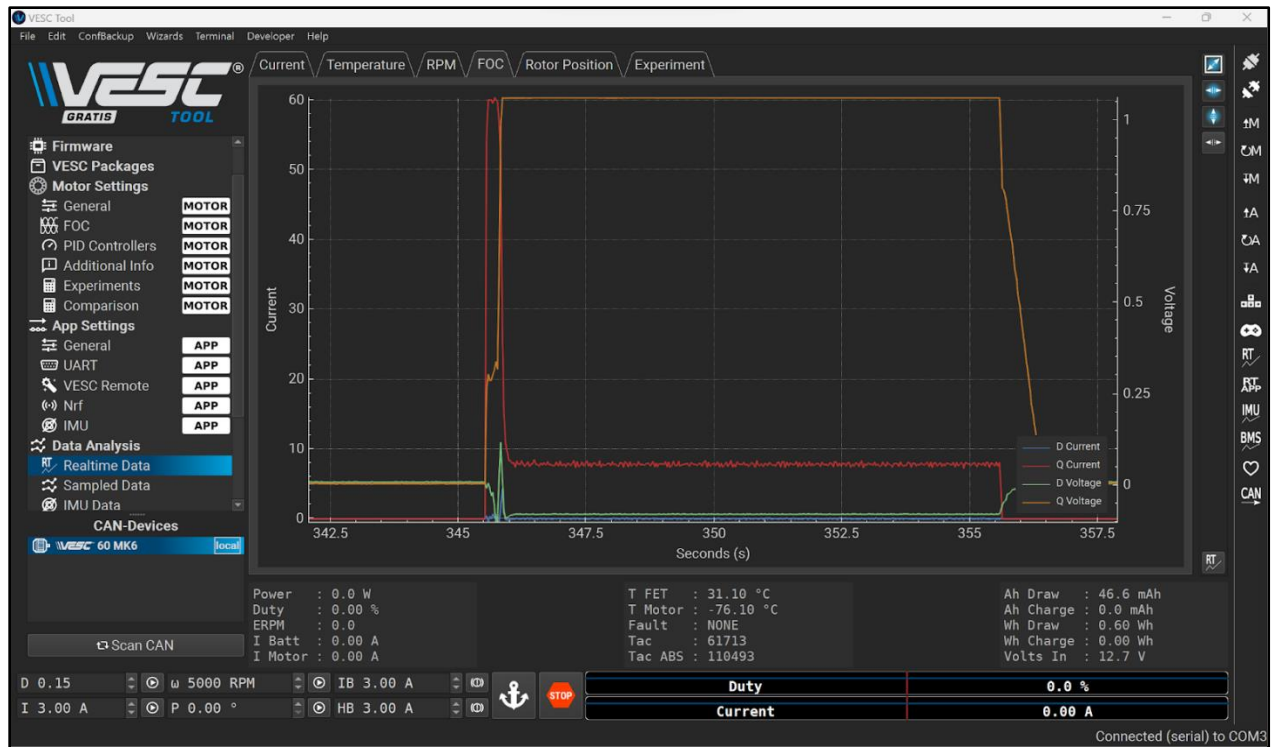


Figure 29. 0.15 Duty Cycle: FOC Graph

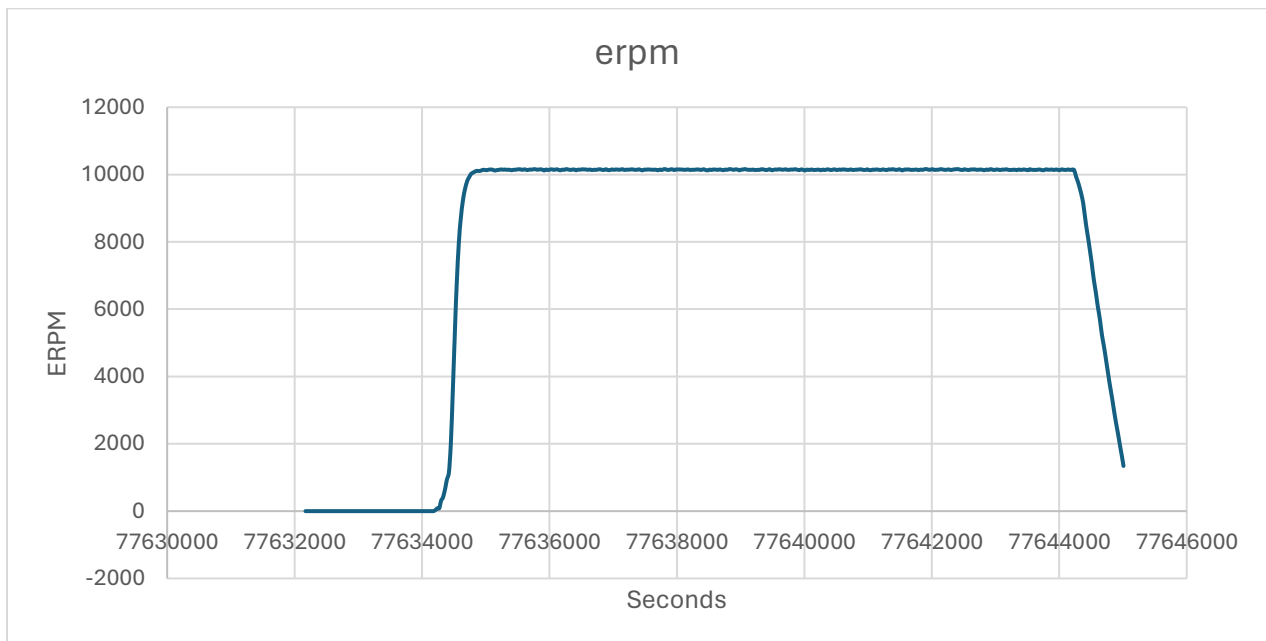


Figure 30. 0.15 Duty Cycle: Excel RPM Graph

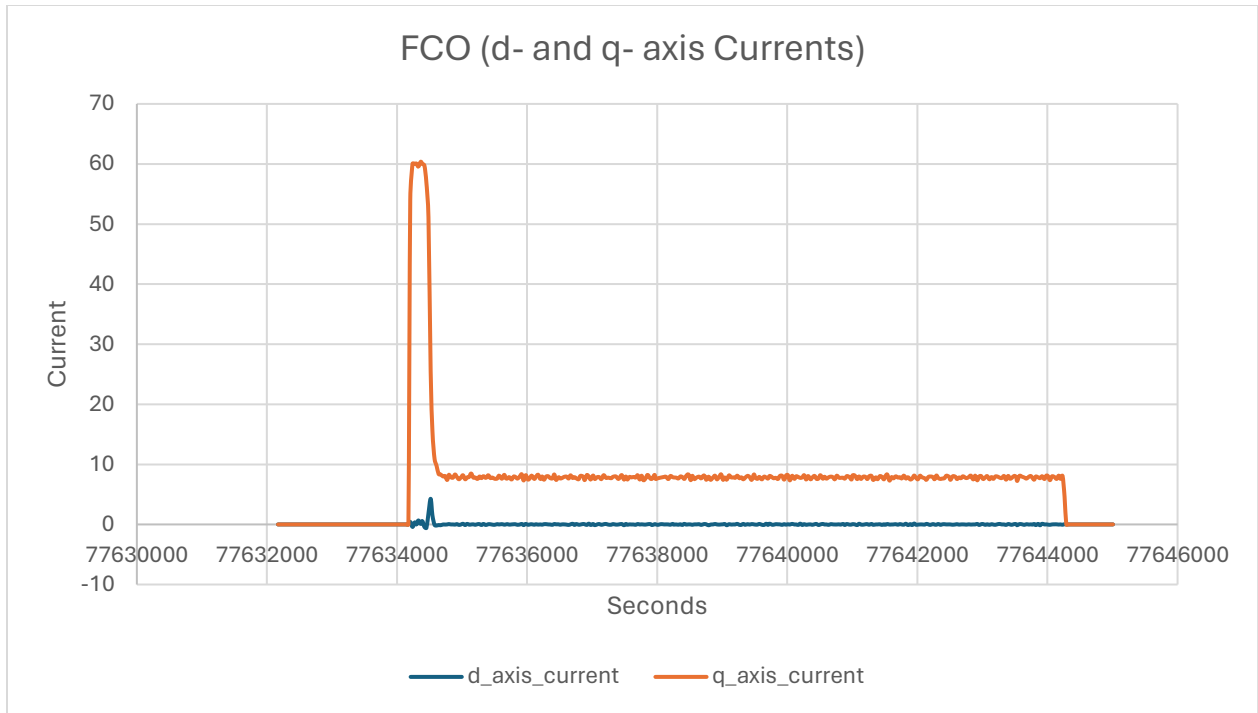


Figure 31. 0.15 Duty Cycle: Excel FOC Current (d- and q-axis Currents) Graph

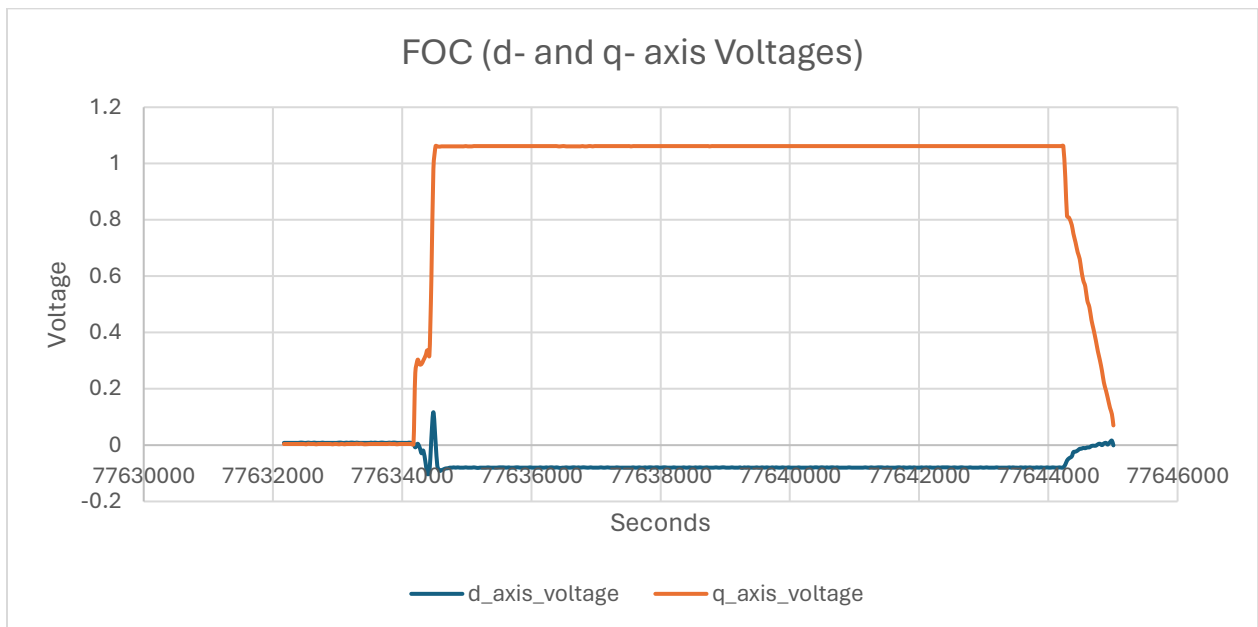


Figure 32. 0.15 Duty Cycle: Excel FOC Voltage (d- and q-axis Voltages) Graph

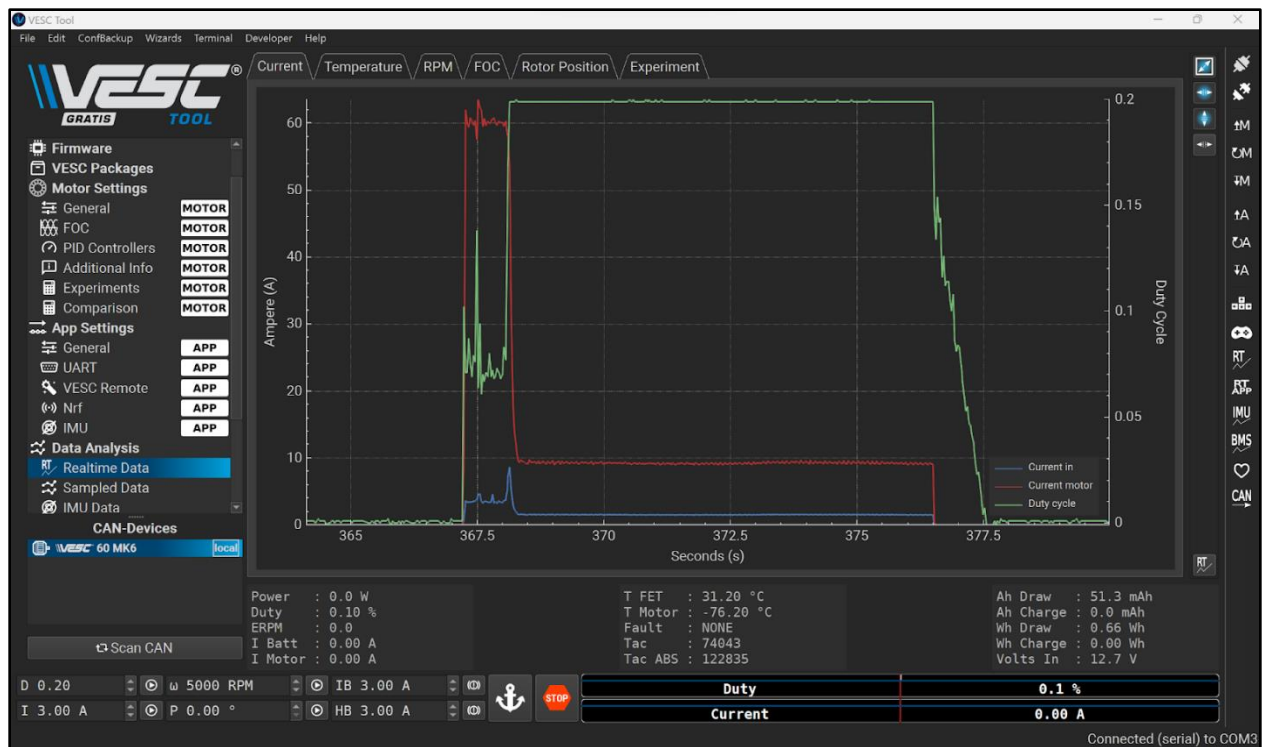


Figure 33. 0.20 Duty Cycle: Current Graph

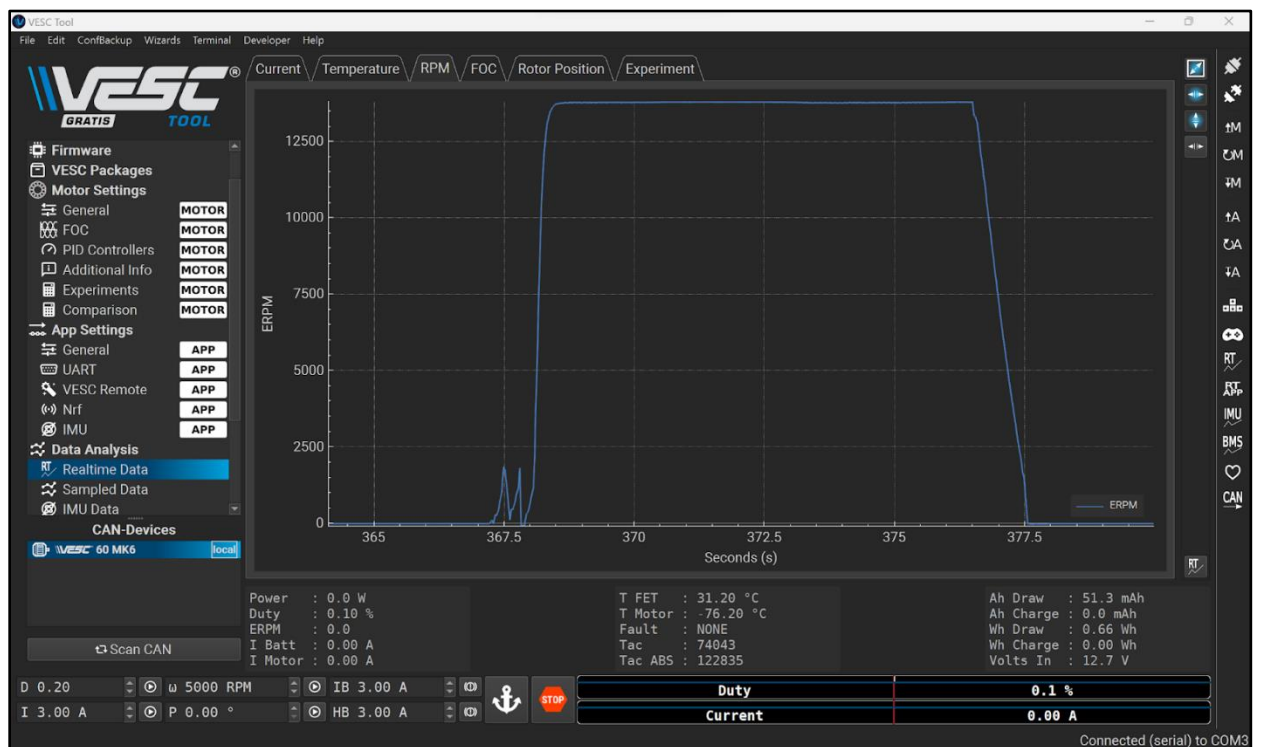


Figure 34. 0.20 Duty Cycle: RPM Graph



Figure 35. 0.20 Duty Cycle: FOC Graph

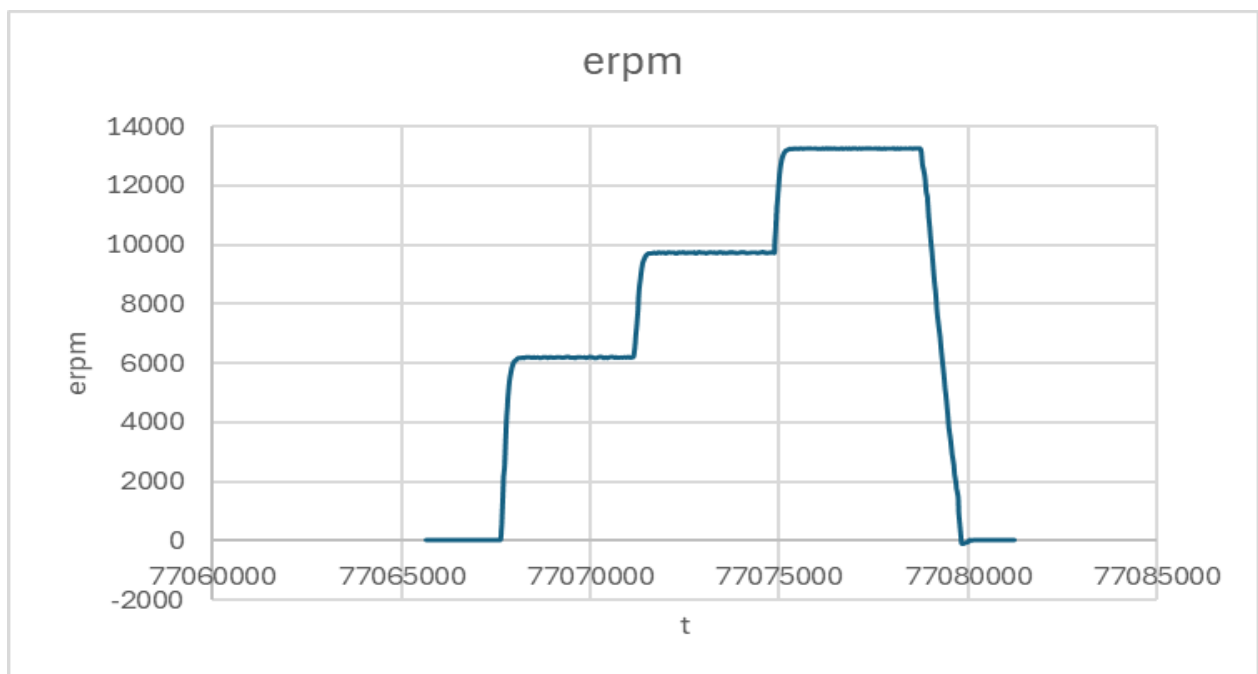


Figure 36. 0.20 Duty Cycle: Excel RPM Graph

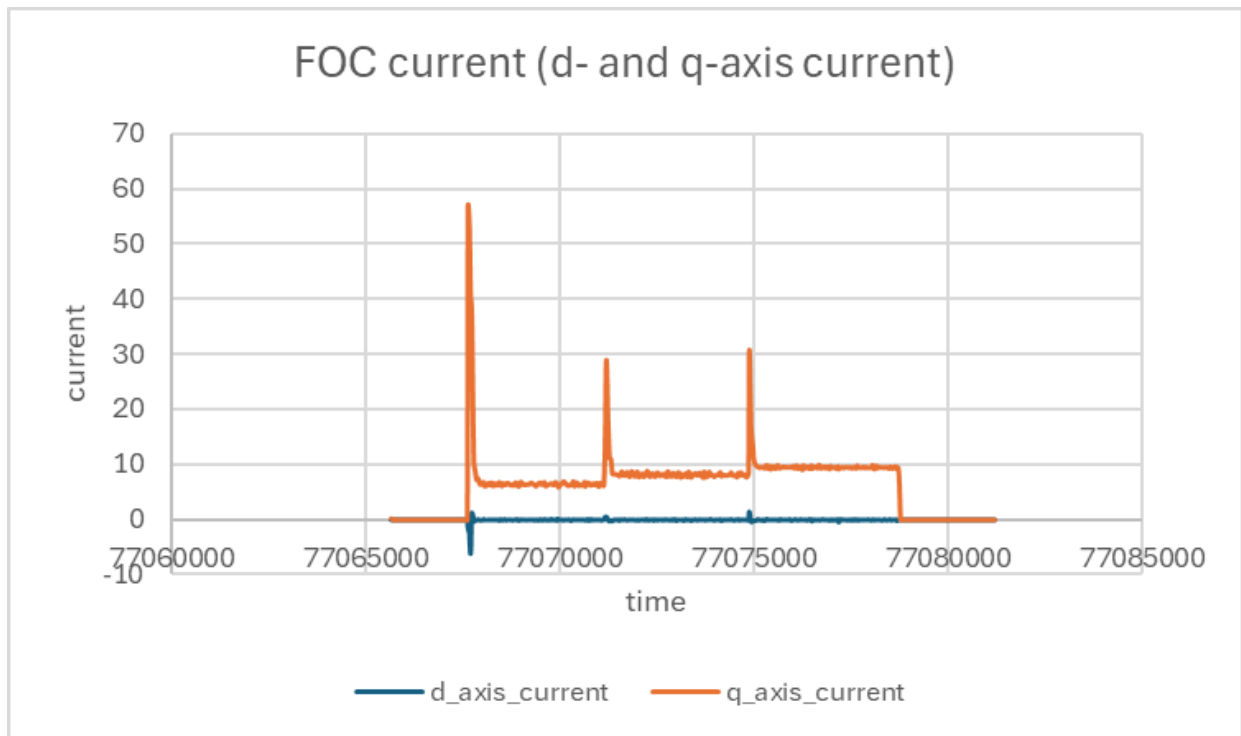


Figure 37. 0.20 Duty Cycle: Excel FOC Current (d- and q-axis Currents) Graph

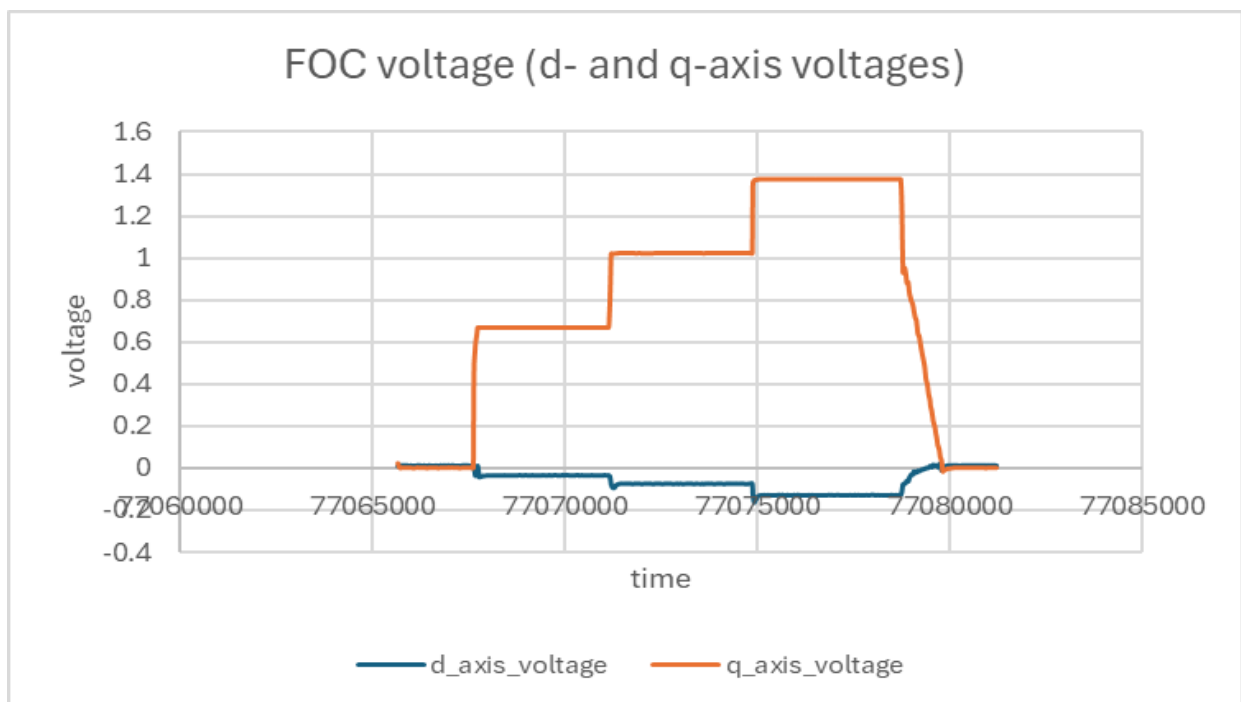


Figure 38. 0.20 Duty Cycle: Excel FOC Voltage (d- and q-axis Voltages) Graph

Question 5

A difference that is observed between the operations with the speed command and the duty cycle command is less noise in the signal when controlling the duty cycle and the duty cycle commands operate at a much higher rpm value. The speed command operation serves as the outer control loop, dynamically regulating the motor's current and voltage to achieve the desired speed setpoint. In contrast, the duty cycle command functions as the inner control loop, directly manipulating the PWM duty cycle to manage motor power. This hierarchical difference in control, coupled with the distinct characteristics of control signals, leads to a potentially more dynamic and noisier response in speed command operation compared to duty cycle command operation. From the observation of rotation speed (RPM), the change of speed was smooth when we changed the duty cycle, but when we changed the speed, there was a sharp change at the command implemented point, and then the speed reduced to the required level, but it still fluctuated around the set value.

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Speed Command: The speed acts as the primary or outer control input, and the motor controller adjusts the current and voltage levels to achieve and maintain the desired speed setpoint. The controller ensures that the motor responds to changes in the speed command, making it the outer loop of control. When examining the real-time waveforms, we observe that the speed command operation exhibits increased noise attributed to the nature of the speed control. Here, the controller iteratively modifies the current and voltage levels to uphold the designated speed setpoint, often leading to fluctuations and noise within the control signals.

Duty Cycle Command: The duty cycle command operates as the secondary or inner loop of control, where the motor controller directly modulates the duty cycle of the PWM signal. This indirectly affects the average voltage applied to the motor. The duty cycle command provides a simpler means of controlling motor power but may lack the precision of speed control. The duty cycle command operation tends to show a smoother control signal. By directly adjusting the average voltage supplied to the motor, the duty cycle ensures more consistent control signals, offering greater stability compared to speed command operation.

Question 6

Both the d- and q-axis represent the flux contribution from the three sinusoidal phase quantities at the same angular velocity. The definition of the d-axis current is that it represents the flux produced by the magnetic field winding. It represents the component of the current that is aligned with the magnetic flux direction. It is the part of the current that is in phase with the magnetic field of the motor produced by the rotor. The d-axis current is kept at zero to maintain a constant magnetic flux and decouple torque and flux control, resulting in efficient motor operation. Under steady-state conditions, the d-axis is usually 0 because there is no change in the magnetic flux linkage in the machine. Thus, if there is no flux to be maintained, there is no need for a current along the axis. The definition of the q-axis current is that it is the axis where the torque is produced. It is the part of the current that is 90° out of phase with the d-axis.

Objective 5: Experiments

Question 7

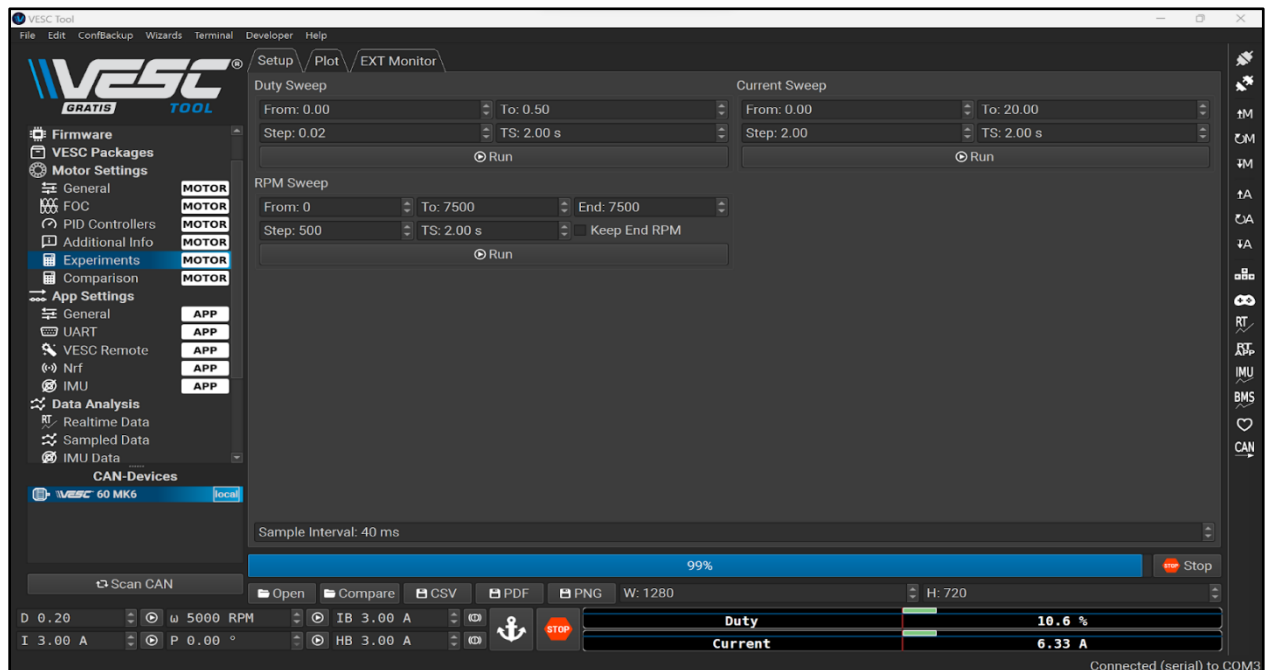


Figure 39. VESC Tool: Data Features

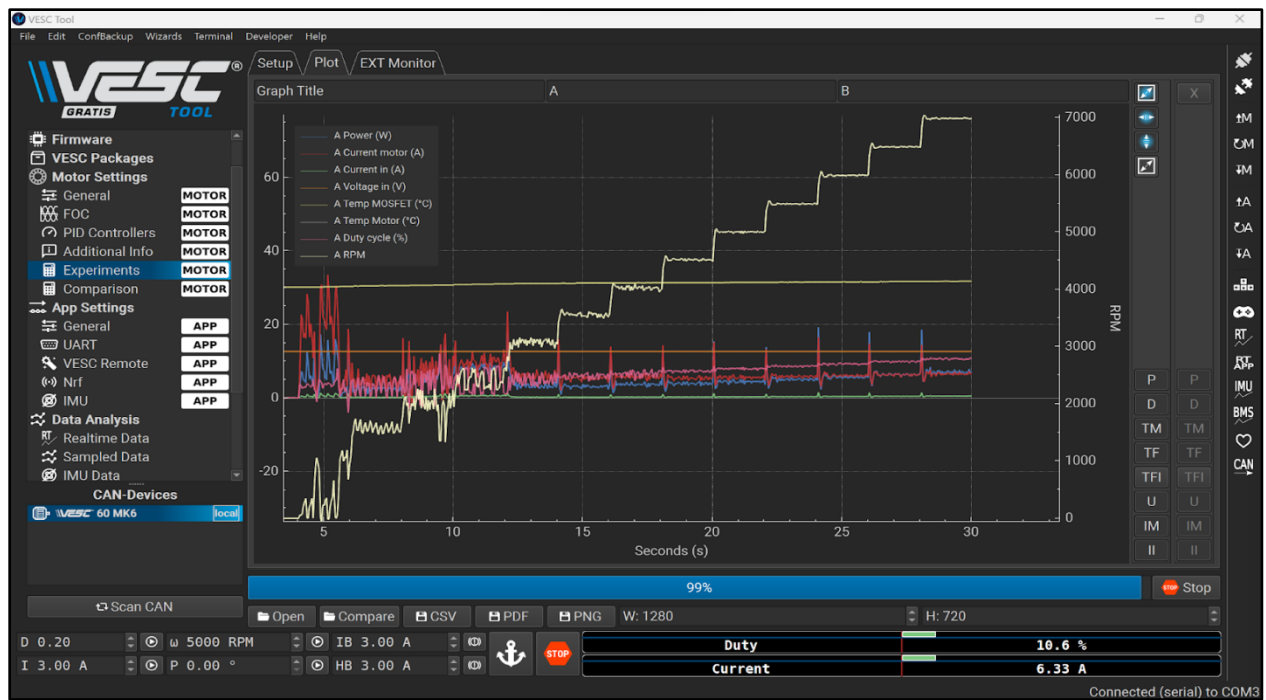


Figure 40. VESC Tool: Plot Tab

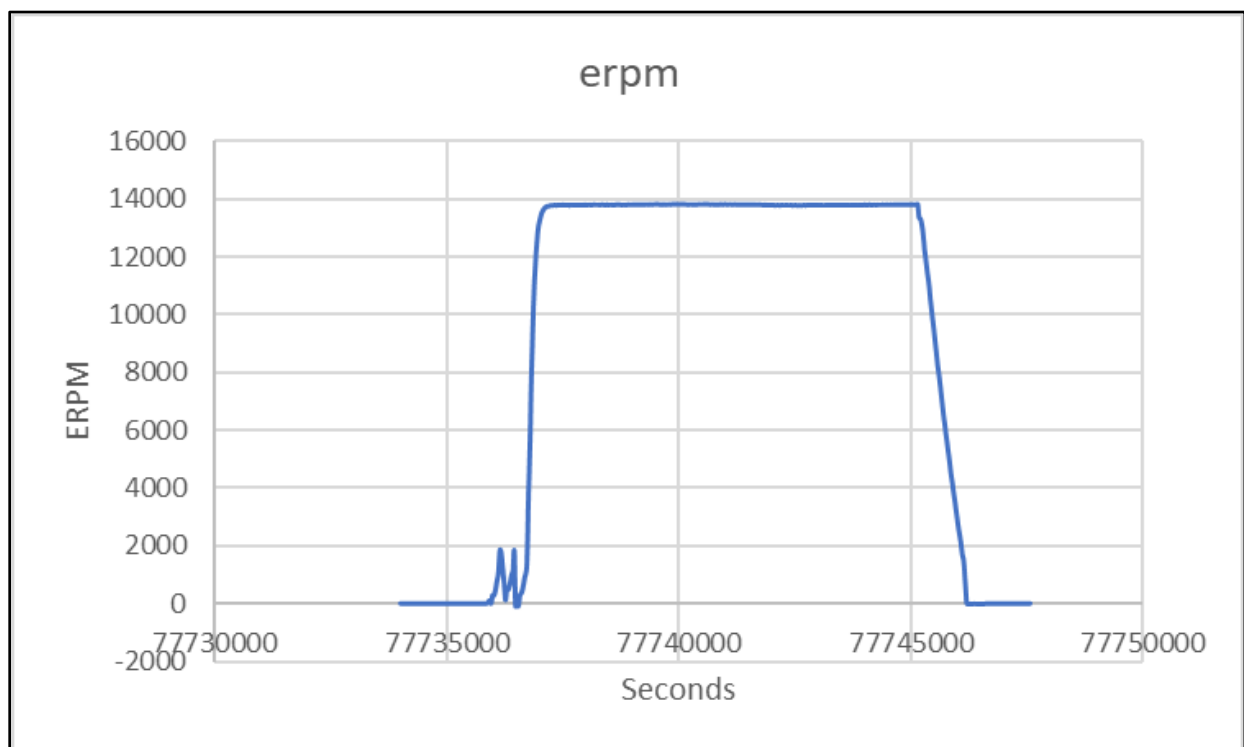


Figure 41. RPM Graph

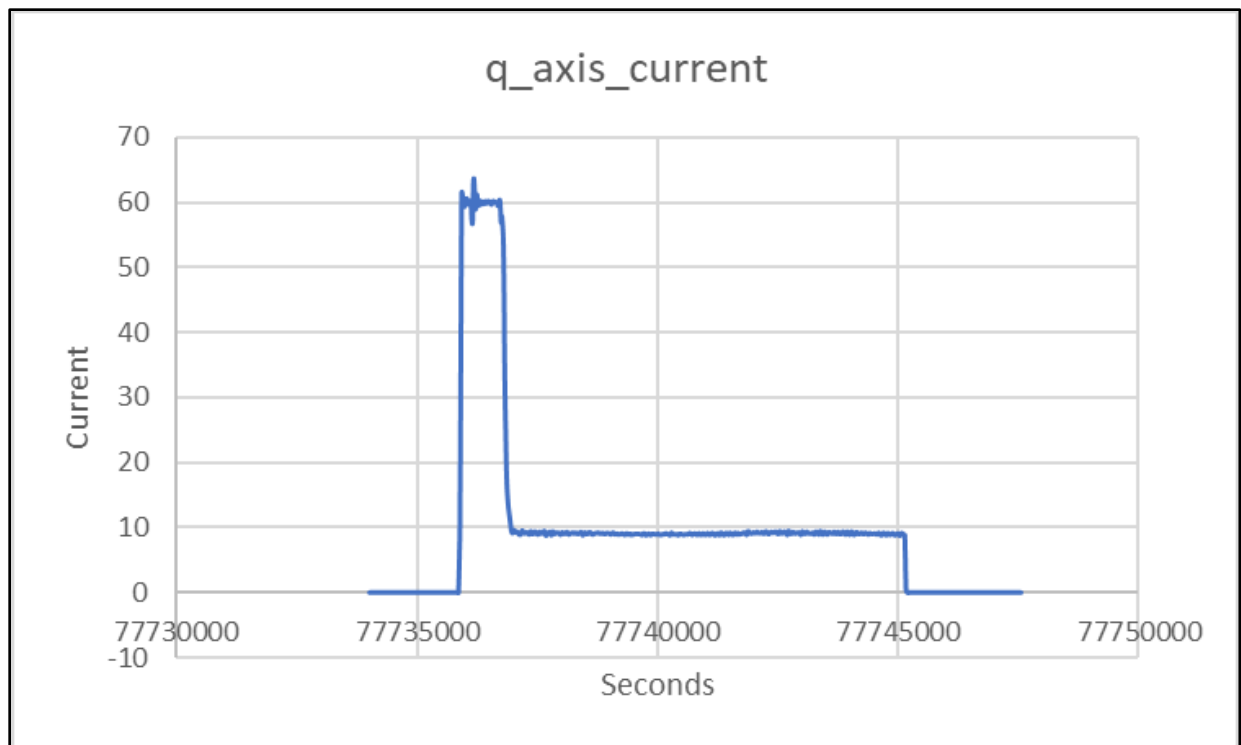


Figure 42. q-axis Current Graph

Objective 6: Record Sampled Data

Question 8



Figure 43. FFT Plot

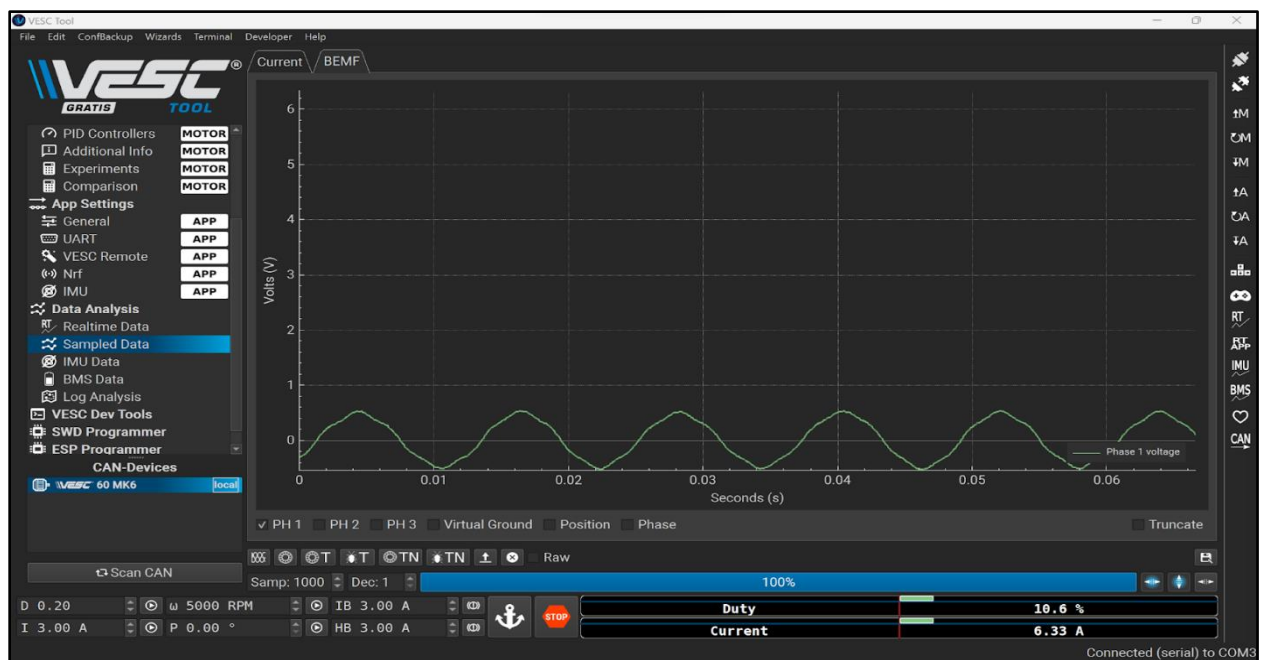


Figure 44. Voltage Plot

Question 9

An FFT plot is a Fast Fourier Transform plot. This plot transforms signals from time domain to frequency domain. It is used when analyzing all types of frequency behaviour. It is a plot of various frequency behaviours and the amplitudes of each frequency signal. It is important to apply the FFT plot to the phase current measurements when analysing the behaviour of a motor because we are able to observe the behaviour of the motor and determine if there are any issues. By doing this, we are able to identify if the motor is operating normally or if any issues or imbalances are occurring that are affecting the performance, operation, health, and status of the motor. By monitoring the change in the current frequency, we can also analyze the acceleration characteristics by viewing the acceleration curve. This will also help to identify any issues impacting the motor. Since the motor we are analyzing is a 3-phase motor, it makes sense to use FFT plots to understand the complex frequency behaviour and performance of the motor. The FFT plots provide a frequency spectrum of the current spectrum, which enables us to see the harmonic components. This is why noise analysis is also important. This is crucial to be able to diagnose faults or distortions in the motor by detecting nonlinear behaviours in the frequencies.